

Volatility Surfaces and Expected Option Returns^{*}

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Abstract

This paper investigates whether deep learning techniques applied to the volatility surface of individual equity options can uncover return predictability arising from retail demand pressure under intermediary constraints. I use a convolutional neural network (CNN) to extract signals from the implied volatility surface to forecast delta-hedged option returns. Long-short portfolios formed on out-of-sample predictions deliver economically large and statistically significant return spreads that survive the impact of transaction costs. Predictability is strongest when financial intermediaries are constrained and it is concentrated in contracts favored by retail investors, consistent with compensation for liquidity provision. The CNN's predictive power cannot be explained by other option characteristics, firm-level predictors, or systematic risk exposure. Feature importance analysis indicates that the model relies on regions of the surface where retail activity is known to be most pronounced.

JEL Classification: G11, G12, G13, C45, C53

Keywords: option return predictability, volatility surface, machine learning

1. INTRODUCTION

Trading in individual equity options has grown dramatically over the past two decades, now exceeding one third of underlying stock volume.¹ This substantial growth has been accompanied by a shift away from the previous dominance of sophisticated investors. Today, retail traders constitute the primary clientele in single-name options, while market-making intermediaries continue to supply liquidity amid tightening capital constraints (Bryzgalova, Pavlova, and Sikorskaya, 2023). Despite the rising importance of this asset class, our understanding of how investor demand shapes expected returns in single-name options remains limited.

This paper investigates whether the interaction between retail demand pressure and the risk-bearing capacity of financial intermediaries gives rise to return predictability in single-name options. When retail investors aggressively demand certain options, intermediaries typically absorb the opposite side of these trades. Because intermediaries face capital constraints and cannot perfectly hedge their inventories (Garleanu, Pedersen, and Poteshman, 2009), they require compensation for warehousing risk. This implies that options subject to strong retail demand should earn lower subsequent returns, particularly when intermediary capital is tight. The implied volatility surface naturally lends itself to study the combined effect of supply and demand conditions, as it aggregates option prices across strikes and maturities. Previous studies have demonstrated that demand pressure affects particular regions of the volatility surface (e.g., Bollen and Whaley, 2004; Eaton, Green, Roseman, and Wu, 2024). This paper takes the reverse approach, using the entire structure of the volatility surface to detect the footprints of investor demand.

Extracting signals from the high-dimensional surface is challenging because it contains many correlated points that may be related to returns through complex and nonlinear interactions. Furthermore, it is unclear ex ante which features contain relevant information or how they should be engineered. This paper aims to circumvent these issues by using a convolutional neural network (CNN, LeCun, 1989), a deep learning model designed for processing data with a grid-like structure (Goodfellow, Bengio, and Courville, 2016). CNNs are widely used in image recognition and have recently been applied to the U.S. stock market (Jiang, Kelly, and Xiu, 2023; Kelly, Kuznetsov, Malamud, and Xu, 2023; Murray, Xia, and Xiao,

¹According to data collected from CRSP and the Options Clearing Corporation (OCC), which is visualized in Figure A.1.

2024). Intuitively, the CNN slides a kernel (or “filter”) over the input matrix, allowing it to detect shapes (or “edges”). These filters are automatically learned and can capture nuanced patterns from raw data. This allows the model to uncover subtle demand-driven patterns without requiring hand-crafted predictors.

The empirical design is as follows. For each U.S. common stock, the CNN takes the two-dimensional volatility surface as input and generates a return forecast for an at-the-money call option that expires in the subsequent month. Following Zhan, Han, Cao, and Tong (2022), I focus on delta-neutral call writing, where the short position in an at-the-money call option is hedged with a long position of delta shares in the underlying stock.² This design abstracts from directional stock predictability and isolates option-specific mispricing, which is central for detecting demand-driven return variation (Broadie, Chernov, and Johannes, 2009). Forecasts are generated in a strictly out-of-sample setting using an expanding-window scheme. The sample spans January 1996 to December 2022 and covers more than 430,000 option-month observations from OptionMetrics. Following Bryzgalova et al. (2023), I capture retail trading activity by identifying trades that are executed through a single-leg price improvement mechanism. Finally, intermediary constraints are proxied by the intermediary capital ratio of He, Kelly, and Manela (2017).

The empirical analysis begins by demonstrating that the shape of the volatility surface is significantly related to expected option returns. A natural way to test this is by using out-of-sample CNN forecasts as a signal to sort delta-hedged options into decile portfolios. The resulting dollar open interest (equal) weighted long-short portfolio generates economically large and statistically significant return spreads of 2.77% (2.01%) per month with an annualized Sharpe ratio of 1.47 (3.20). Importantly, this performance cannot be fully explained by the observation that single-name options tend to have high average bid-ask spreads. The average returns of the strategy remain economically and statistically significant even after accounting for transaction costs of up to 100% of the quoted bid-ask spread.

I proceed to provide evidence to support the idea that the variation in average returns

²I adopt a number of measures to ensure a realistic investment process and to deal with common microstructure issues in empirical option research. First, I focus on at-the-money call options to avoid measurement errors in prices (Duarte, Jones, and Wang, 2024). Second, I apply data filters only at portfolio formation to avoid potential look-ahead bias (Duarte, Jones, Mo, and Khorram, 2025). Third, I hold option positions until expiration and do not rebalance delta hedges to reduce transaction costs. Fourth, I weight positions by dollar open interest to alleviate the impact of thinly traded options (e.g., as in Zhan et al., 2022; Bali, Beckmeyer, Moerke, and Weigert, 2023).

across CNN deciles is related to demand pressure. Using a subsample analysis based on the intermediary capital ratio, I show that profitability is concentrated in periods when financial intermediaries are constrained. This suggests that the CNN portfolio is rewarded for countercyclical liquidity provision (Gromb and Vayanos, 2002). To answer the question of who the CNN provides liquidity to, I show that CNN forecasts are strongly related to retail activity. Specifically, higher expected returns to delta-neutral call writing are accompanied by a statistically significant increase in the retail share of total option trading volume. Taken together, this evidence supports the hypothesis that the CNN strategy earns high returns by supplying liquidity to retail investors, particularly when intermediaries are constrained.

I use Fama and MacBeth cross-sectional regressions to address two concerns regarding the CNN model. First, the model may simply capture existing sources of option return predictability documented in the literature. Second, specific shapes of the surface may reflect structural departures from the Black and Scholes model rather than demand pressure. For example, stochastic volatility and jump risk can create deviations from the flat volatility surface implied by Black and Scholes (1973). To address these questions, I regress realized option returns on CNN predictions while controlling for a comprehensive set of option-implied characteristics. These include measures such as volatility of volatility and the term structure of implied volatility, which may proxy for the presence of stochastic volatility. I also include skewness and kurtosis, which are often used to capture jump risk. The results show that the model cannot be explained by these variables and that it subsumes the predictive power of several measures, including risk-neutral variance, the slope of the implied volatility term structure (Vasquez, 2017), realized variance, and the variance risk premium (Bakshi and Kapadia, 2003). Despite this, less than half of the cross-sectional variation in the model’s predictions can be explained by these conventional surface-based characteristics. In addition, the predictive ability of the CNN cannot be explained by previously documented firm-level predictors such as profitability and cash holdings (Zhan et al., 2022).

Another concern is that the CNN might detect patterns that proxy for systematic risk exposures. To address this, I estimate factor models with latent factors and time-varying loadings based on a large set of stock and option characteristics (Kelly, Pruitt, and Su, 2019; Goyal and Saretto, 2025). The results show that the CNN based investment strategy attains large and statistically significant alphas, reflecting abnormal returns rather than compen-

sation for taking on systematic risk. In addition, I benchmark CNN predictions against forecasts of a conventional fully connected neural network and conditional expected returns of the latent factor model. These alternative models explain little of the CNN portfolio’s variation, with R^2 values below 12%, highlighting the ability of the CNN to extract novel information from the volatility surface.

Interpreting what the CNN learns provides an additional test of the hypothesis that demand pressure leaves identifiable footprints in the volatility surface. While machine learning models are often difficult to interpret due to their nonlinear transformations of input data, I shed light on the CNN’s forecasts in two ways. First, Shapley values (Lundberg and Lee, 2017) reveal that at-the-money calls with one month until expiration, and deep out-of-the-money puts are the most influential regions of the volatility surface for predicting option returns. The prominence of short-dated options and at-the-money calls is consistent with prior evidence that retail investors concentrate their activity in these contracts, favor calls over puts, and trade heavily around the money (Bryzgalova et al., 2023; Bogousslavsky and Muravyev, 2024; Eaton et al., 2024). The important role of deep out-of-the-money puts is likewise in line with evidence that retail investors exhibit a preference for lottery-like payoffs (Boyer and Vorkink, 2014; Filippou, Garcia-Ares, and Zapatero, 2018). Second, visualizations of volatility surfaces associated with high and low CNN forecasts reveal similar intuitive patterns that are easily recognizable and closely aligned with previously documented predictors.

Finally, the paper provides a number of additional robustness checks to confirm that the results are not artifacts of liquidity effects, microstructure noise, or overfitting. Specifically, I show that the CNN also predicts returns of options that are hedged daily rather than monthly. Furthermore, sorting alternative instruments (e.g., straddles) into CNN deciles, yields a profitable long-short portfolio as well. In contrast, repeating this exercise for the underlying stock does not generate a significant return spread. Taken together, my results suggest that the CNN portfolio uncovers demand-driven return variation rather than directional stock price movements.

Related literature. My research contributes to the existing body of literature on the cross-section of equity option returns. Studies in this field typically analyze returns of option portfolios formed by sorting on some observable characteristic (e.g., Goyal and Saretto,

2009; Bali and Murray, 2013; Cao and Han, 2013; Boyer and Vorkink, 2014; Vasquez, 2017; Christoffersen, Goyenko, Jacobs, and Karoui, 2018; Hu and Jacobs, 2020; Zhan et al., 2022; Heston, Jones, Khorram, Li, and Mo, 2023). My approach departs from this literature by constructing portfolios based on deep learning signals.

Recently, an emerging strand of the literature has started to employ machine learning methods in empirical options research. Büchner and Kelly (2022) present a factor model with time-varying betas and latent risk factors which accounts for more than 85% of the variation in index option returns. Goyal and Saretto (2025) extend the model to single-name options and document that most option trading strategies can be explained by systematic risk factors.³ My paper shows that the CNN model generates abnormal returns relative to this factor model. Finally, Bali et al. (2023) use a large set of self-constructed characteristics to predict option returns with machine learning. My paper differs from this work in that the CNN automatically constructs features from raw data rather than using prespecified signals.

More generally, my work is related to the growing literature on machine learning in empirical asset pricing.⁴ Notable papers in this field have pioneered the use of CNNs to predict stock returns from price charts (Jiang et al., 2023; Murray et al., 2024) and the volatility surface (Kelly et al., 2023). My paper is fundamentally different from these papers in that I focus on delta-hedged option returns rather than stock returns.

Finally, my paper is also connected to the theoretical option pricing literature (Black and Scholes, 1973; Heston, 1993; Duffie, Pan, and Singleton, 2000; Carr and Wu, 2004). While parametric no-arbitrage models are easier to interpret than deep learning models, they have not been successful in accurately describing the variation in delta-hedged option returns (Israelov and Kelly, 2017).

The remainder of this paper proceeds as follows. In Section 2, I outline the CNN architecture and training process. Section 3 describes the data used to estimate the model and Section 4 presents my main empirical findings. Section 5 reports robustness tests and additional results. I conclude in Section 6.

³For further work on the factor structure of option returns, see Horenstein, Vasquez, and Xiao (2022) and Fournier, Jacobs, and Orłowski (2024). Käfer, Mörke, and Wiest (2023) show that option factors exhibit momentum.

⁴For a comprehensive review of this literature see Kelly, Xiu, et al. (2023).

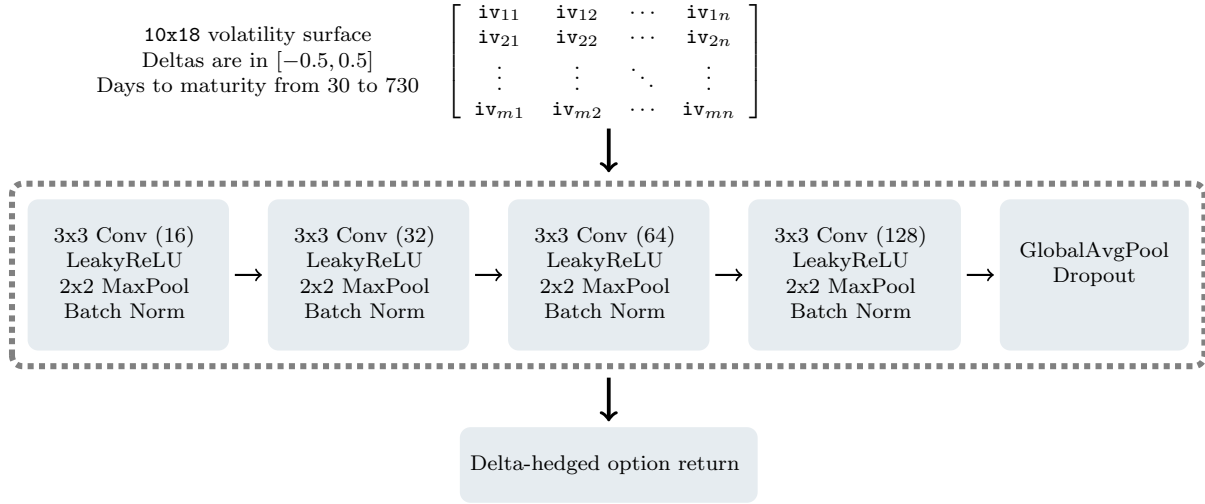


Figure 1: Diagram of CNN model architecture. This figure visualizes the layers of my CNN model. It takes as an input the 10x18 volatility surface matrix from OptionMetrics, consisting of out-of-the-money call and put options with deltas ranging from -0.5 to 0.5 and time to maturity equal to 30, 60, 91, 122, 152, 182, 273, 365, 547 or 730 days. The model consists of three convolutional layers, each with a 3x3 filter, a nonlinear activation function, maximum pooling operation, and batch normalization. The linear output layer is preceded by global average pooling and dropout.

2. CNN MODEL

This section briefly describes the architecture of the CNN model (Section 2.1) as well as how it is regularized and trained (Section 2.2).

2.1. Architecture and regularization

My choices regarding the model architecture are similar to those in previous papers (Gu, Kelly, and Xiu, 2020; Jiang et al., 2023; Kelly et al., 2023). I stack four convolutional blocks, each containing a convolutional layer, the first with 16 filters, the second with 32, the third with 64, and the fourth with 128 filters (LeCun, 1989). Each filter has a kernel size of three, and moves across the image with a stride of two. To preserve the initial input size, I add zero padding to each image. The nonlinear activation function is LeakyReLU, i.e. a rectified linear unit with a small slope for negative values to prevent the “dying ReLU” problem (Maas, Hannun, Ng, et al., 2013). I apply a maximum pooling operation after each convolutional layer, as is common practice (Zhou and Chellappa, 1988). The pooling layer also has a kernel size and stride size equal to two. The final model is summarized in Figure 1.⁵

⁵I implement the model using PyTorch (Paszke, Gross, Massa, Lerer, Bradbury, Chanan, Killeen, Lin, Gimelshein, Antiga, et al., 2019).

The goal of the neural network is to predict the percentage delta-hedged return of option i in $t + 1$ using the volatility surface matrix $\mathbf{iv}$ at time t . Akin to Gu et al. (2020), the corresponding prediction error model reads

$$\begin{aligned} r_{i,t+1}^{dh} &= \hat{r}_{i,t+1}^{dh} + \varepsilon_{i,t} \\ \hat{r}_{i,t+1}^{dh} &= g(\mathbf{iv}_{i,t}; \boldsymbol{\theta}) \end{aligned} \tag{1}$$

where $g(*)$ is a flexible nonlinear function and $\boldsymbol{\theta}$ is a set of parameters. Model parameters are estimated to minimize the mean squared loss of the training sample

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{NT} \sum_{t=1}^T \sum_{i=1}^N (r_{i,t+1} - \hat{r}_{i,t+1}^{dh})^2. \tag{2}$$

The total number of parameters in $\boldsymbol{\theta}$ is large and amounts to 97,761. Hence, heavy regularization is required to guard against overfitting and ensure out-of-sample generalizability. I adapt similar optimization and regularization procedures as in previous studies (e.g. Gu et al., 2020; Jiang et al., 2023). The first regularization technique is batch normalization, which standardizes the output of each convolutional block (Ioffe and Szegedy, 2015). Further, I apply 50% dropout before the linear output layer (Srivastava, Hinton, Krizhevsky, Sutskever, and Salakhutdinov, 2014). As a last regularization technique, I use an ensemble approach (Hansen and Salamon, 1990; Dietterich, 2000). More specifically, I train an ensemble of 20 models, each initialized with a different random seed, and take an average across their predictions. Optimization is carried out using Adam, which typically converges much faster than regular stochastic gradient descent (Kingma and Ba, 2014). I choose a learning rate of 1×10^{-2} and a batch size of 1024. Weights are randomly initialized using Xavier uniform (Glorot and Bengio, 2010).

2.2. Training and out-of-sample design

Because neural networks are prone to overfitting, I evaluate the model exclusively on out-of-sample predictions. Therefore, I split the sample into 7 years of training (1996-2002) and 20 years of testing (2003-2022). I implement an expanding-window procedure in which the model is initially trained on the first seven years of data for ten epochs and then used to generate

forecasts for the following year (Kelly et al., 2023).⁶ The training window is subsequently expanded to include this new year, and the model is retrained for an additional five epochs before producing the next set of forecasts. Repeating this procedure yields 20 years of strictly out-of-sample return predictions. To avoid data snooping and reduce computational burden, I do not conduct hyperparameter tuning on a validation set, but instead adopt parameters choices that have worked well in previous studies.

3. DATA AND DESCRIPTIVE STATISTICS

I now proceed to outline my sample and provide details regarding key variables. Section 3.1 describes the data sources and sample construction. Section 3.2 introduces the return measures, and Section 3.3 explains how implied volatility surfaces are obtained.

3.1. Data sources and filters

The primary data source is OptionMetrics IvyDB, from which I obtain historical prices of single-name options on U.S. common stocks and their interpolated implied volatility surface. I use the option price file to draw closing best bid and offer price, trading volume, open interest, and Greeks for each option. The security price file provides closing prices and cumulative adjustment factors of the underlying. Underlying stock characteristics are collected from Chen and Zimmermann (2022) and the risk-free rate is taken from Ken French’s Data Library.⁷ The sample period spans January 1996 to December 2022.

I follow the standard practice of applying a series of data filters to remove erroneous prices and options with relatively low trading activity (Goyal and Saretto, 2009; Cao and Han, 2013; Zhan et al., 2022). Contrary to some previous studies, I use only information available at portfolio formation to eliminate these observations.⁸ First, I only keep options written on U.S. common stocks (CRSP share codes 10 or 11) with a share price above 5 USD.⁹ Second, I discard options with non-standard settlement date, irregular expiry, or AM

⁶I shuffle all observations during each training step. Furthermore, each grid point of the surface is normalized with mean and standard deviation of all values in the initial training set.

⁷See https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

⁸This is to avoid look-ahead bias, which can lead to spurious option return predictability and inflated Sharpe ratios (Duarte et al., 2025).

⁹I link OptionMetrics’ *secid* to CRSP’s *permno* using the `opcrcsphist` linking table provided by WRDS. I only retain links with the highest quality (score equal to 1).

settlement. Third, I require that the last trade date matches the portfolio formation date. Fourth, I filter out options with trading volume or open interest equal to zero. Fifth, I require that the mid price does not violate model-free arbitrage bounds and that the bid price is greater than zero and below the ask. Last, I remove options with missing delta.

3.2. *Delta-hedged option returns*

The main variable of interest is the return from writing a call option that is hedged with a long position in the underlying stock. Following the literature, I study delta-hedged rather than naked option positions (e.g., [Zhan et al., 2022](#)). Delta-hedging serves two purposes. First, it eliminates predictability inherited from the underlying stock. Second, it produces return distributions with more favorable statistical properties, in contrast to raw option returns that often exhibit fat left tails at -100% . I focus on call options, which are more actively traded than puts ([Christoffersen et al., 2018](#)).

Following [Büchner and Kelly \(2022\)](#), I study monthly investment periods that begin immediately after each major option expiration and end on the subsequent expiry date, typically the third Friday of the month. To mitigate microstructure biases in option returns ([Duarte et al., 2024](#)), I restrict the sample to at-the-money contracts, since deep out-of-the-money options typically exhibit wide bid-ask spreads and deep in-the-money options often violate no-arbitrage conditions. Consequently, for each underlying, among all options expiring next month with moneyness between 0.8 and 1.2, I select the contract closest to being at-the-money. In case there remain multiple contracts, I prefer the one that is out-of-the-money.¹⁰ The final sample comprises 436,152 option-month observations. The time-series evolution of the sample size is reported in [Appendix A.2](#).

I compute the excess return to delta-neutral call writing following [Goyal and Saretto \(2009\)](#) and [Zhan et al. \(2022\)](#)

$$r_{t+1}^{dh} = \frac{\Delta_t \cdot S_{t+1} - \max(S_{t+1} - K, 0)}{\Delta_t \cdot S_t - C_t} - 1 - r_t^f, \quad (3)$$

where Δ denotes the BSM option delta, S the underlying price, K the strike price, and C the

¹⁰In some rare cases, multiple options share the same underlying, strike price, and expiration date. Following [Bali and Murray \(2013\)](#) I exclude these observations.

	Mean		Sd		Min	Q1	Q2	Q3	Max	Count
	OIW	EW	OIW	EW						
Option return (%)	0.35	0.48	8.73	10.96	-120.92	-3.38	1.58	6.10	41.17	1,347.26
Stock return (%)	0.81	1.01	11.56	12.49	-60.54	-5.56	0.68	7.09	96.19	1,347.26
Days to maturity	28.04	28.04	0.00	0.00	28.04	28.04	28.04	28.04	28.04	1,347.26
Moneyness (S/K)	1.00	0.99	0.03	0.05	0.81	0.96	0.99	1.02	1.19	1,347.26
Delta	0.54	0.49	0.10	0.14	0.08	0.40	0.50	0.59	0.96	1,347.26
Gamma	4.24	3.71	1.80	1.78	0.64	2.49	3.37	4.60	21.95	1,347.26
Vega	0.10	0.10	0.01	0.01	0.02	0.10	0.11	0.11	0.11	1,347.26
Theta	-0.32	-0.34	0.15	0.15	-1.41	-0.42	-0.31	-0.23	-0.04	1,347.26
Quoted spread (%)	7.95	23.66	8.61	24.59	0.97	8.59	15.23	28.80	166.14	1,347.26

Table I: Periodic cross-sectional summary statistics. This table presents time-series averages of cross-sectional summary statistics for the sample of at-the-money call options. It shows the average mean, standard deviation (Sd), minimum (Min), 25th percentile (Q1), median (Q2), 75th percentile (Q3), and maximum (Max) of the distribution of delta-hedged option returns, stock returns, calendar days to expiration, moneyness, delta, gamma, vega, theta, and quoted spread. Gamma is multiplied by S/100 to reflect the response to a 1% stock price move. Vega and theta are both scaled by the underlying stock price. OIW (EW) denotes the dollar open interest (equal) weighted average and standard deviation. The column labeled “Count” indicates the average number of options for which each variable is available. Data are monthly from 1996 to 2022.

call price.¹¹ Positions are held until expiration to minimize transaction costs and to avoid using quotes of options with zero open interest or trading volume. Furthermore, I do not rebalance delta hedges to reduce transaction costs, which is in line with Goyal and Saretto (2009), Bali and Murray (2013), Zhan et al. (2022), and Goyal and Saretto (2025).¹²

I summarize the final cross-section of options in Table I. The average month spans 1,347 calls with 28 days to maturity and moneyness close to 1. The equal weighted average quoted spread is 23.7%, but falls to 8.0% when portfolios are weighted by dollar open interest. To alleviate the influence of illiquid or thinly traded options, I report main results using dollar open interest weighted portfolios (Duarte et al., 2024). The average equal weighted (dollar open interest weighted) delta-hedged call return is 0.48% (0.35%) per month, in contrast to the zero expected return implied by the BSM model.

3.3. Implied volatility surfaces

The volatility surface data are obtained from OptionMetrics, which provides interpolated implied volatilities for each underlying on each day.¹³ OptionMetrics employs a proprietary

¹¹Computing returns this way ignores the possibility of early exercise. However, it is unlikely to affect my results as early exercise is rarely optimal for options close to the money (Heston et al., 2023).

¹²In Section 5.1, I report results for daily delta hedging as well.

¹³https://wrds-www.wharton.upenn.edu/documents/1504/IvyDB_US_v5.4_Reference_Manual.pdf.

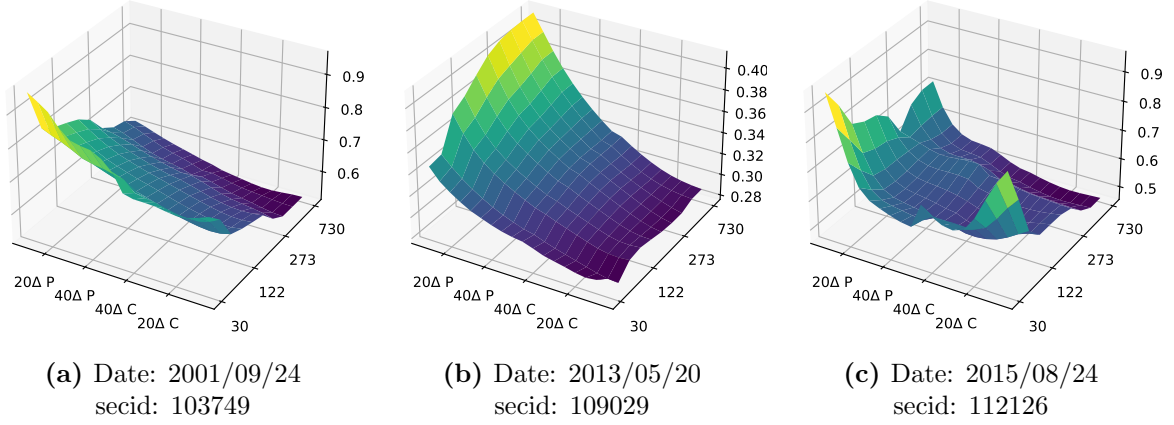


Figure 2: Exemplary volatility surfaces. This figure shows three randomly drawn volatility surfaces which plot implied volatility against option delta and calendar days to expiration.

Cox-Ross-Rubinstein binomial tree model to compute implied volatilities and then interpolates values across grid points using a kernel smoother. A surface is reported only when sufficient option price data are available to support reliable interpolation, which is the case for 99.999% of all stock-month observations in my sample. Following [Bakshi, Kapadia, and Madan \(2003\)](#) and [Martin and Wagner \(2019\)](#), I merge out-of-the-money calls and puts at the money to obtain a single surface for each underlying.¹⁴ The resulting surface is represented as a 10×18 matrix, with maturities ranging from 30 to 730 days and deltas spanning -0.5 to 0.5. Figure 2 illustrates representative examples and highlights the variety of shapes that can be observed in the data.

4. EMPIRICAL RESULTS

The following sections present the main empirical results. Section 4.1 documents the returns to portfolios based on CNN predictions, reports portfolio characteristics, and analyzes performance during periods of economic stress and intermediary constraints. Section 4.2 evaluates the impact of transaction costs, and Section 4.3 links CNN predictions to retail trading activity. Section 4.4 examines the relation between CNN forecasts and commonly studied option characteristics. Section 4.5 shows that the CNN portfolio generates abnormal returns relative to a factor model, while Section 4.6 compares its performance to compet-

¹⁴The reason for this practice is that in-the-money options are likely to be less informative than out-of-the-money options because they trade less frequently, have wider bid-ask spreads, and higher prices due to their intrinsic value which is unrelated to future outcomes ([Birru and Figlewski, 2012](#)).

ing models. Finally, Sections 4.7 and 4.8 interpret CNN forecasts using feature importance measures and visualizations of surfaces associated with high and low predicted returns.

4.1. Machine learning portfolios

I assess the economic value of the CNN model by forming portfolios based on its return forecasts (see Gu et al., 2020; Bali et al., 2023). On the first trading day after each major monthly option expiration, I generate out-of-sample predictions of delta-neutral call writing returns and sort options into decile portfolios. Portfolios are held until the subsequent expiration, and average realized returns are computed over this horizon. The long-short strategy writes options in the highest predicted return decile and buys options in the lowest decile. Following Zhan et al. (2022) and Bali et al. (2023), positions are weighted by dollar open interest at portfolio formation to mitigate the influence of thinly traded contracts.

Table II reports average predicted and realized excess returns, t-statistics, and Sharpe ratios of decile portfolios. Panel (a) displays results for portfolios weighted by dollar open interest. Average predictions tend to be larger than realized returns, especially for the most extreme deciles. Nonetheless, average realized returns in the bottom and top decile are economically large and statistically significant. This is also reflected in the performance of the corresponding zero net investment strategy (row HML), which achieves an average monthly return of 2.8% (t-statistic of 7.8). Its annualized out-of-sample Sharpe ratio amounts to 1.5. In comparison, the equal weighted long-short portfolio (shown in Panel b) earns an average return of 2 percentage points per month and an annualized Sharpe ratio of 3.2.

Ensemble distribution. Because neural networks rely on stochastic training algorithms and random weight initialization, each model typically converges to a different set of parameters and captures distinct features of the data. A standard approach in machine learning is therefore to train an ensemble of models and aggregate their predictions to reduce variance and improve generalizability (Hansen and Salamon, 1990; Dietterich, 2000). To evaluate the benefits of ensembling, I analyze the performance of portfolios formed from individual CNN models as well as from their combination. Table 3 reports the distribution of annualized Sharpe ratios across 20 individual CNN models and the ensemble. Individual estimators yield Sharpe ratios ranging from 0.4 to 1.2, with most clustered around 0.8. In contrast, the ensemble achieves a Sharpe ratio of nearly 1.5, consistent with the gains from ensemble

	Pred.	Real.	t	SR		Pred.	Real.	t	SR
Low	-3.22	-1.04	(-3.03)	-0.56	Low	-2.67	-0.13	(-0.51)	-0.11
2	-0.24	-0.02	(-0.08)	-0.02	2	-0.26	-0.09	(-0.39)	-0.09
3	-0.03	-0.18	(-0.08)	-0.20	3	-0.04	-0.12	(-0.57)	-0.14
4	0.07	-0.00	(-0.01)	-0.00	4	0.07	-0.14	(-0.66)	-0.16
5	0.16	-0.02	(-0.10)	-0.02	5	0.16	0.02	(0.08)	0.02
6	0.26	0.01	(0.03)	0.01	6	0.27	0.05	(0.22)	0.05
7	0.40	0.12	(0.47)	0.11	7	0.40	0.22	(0.96)	0.21
8	0.59	0.47	(1.76)	0.39	8	0.60	0.30	(1.21)	0.27
9	0.95	0.53	(2.08)	0.39	9	0.96	0.62	(2.37)	0.55
High	2.67	1.72	(4.92)	0.94	High	2.57	1.88	(6.16)	1.51
HML	5.89	2.77	(7.75)	1.47	HML	5.24	2.01	(14.17)	3.20

(a) Dollar open interest weighted

(b) Equal weighted

Table II: CNN portfolios. This table shows average excess returns of decile portfolios based on out-of-sample predictions of the CNN model. Panel (a) shows portfolios weighted by dollar open interest at the beginning of the investment period. Panel (b) shows equal weighted portfolios. t -statistics adjusted for heteroscedasticity and autocorrelation are reported in parentheses. Data are at the monthly frequency and span January 2003 to December 2022.

learning documented by [Kelly et al. \(2023\)](#).

Subsample analysis. I next examine the performance of the CNN model across different market conditions to test whether it is related to limits to arbitrage, the funding conditions of financial intermediaries, or investor sentiment. [Avramov, Cheng, and Metzker \(2023\)](#) show that investment strategies based on deep learning signals typically perform well when limits to arbitrage are high, whereas they face difficulties in periods with fewer frictions and more arbitrage activity. This is in line with [Chordia, Subrahmanyam, and Tong \(2014\)](#), who document that anomalies are attenuated when an increase in market liquidity triggers arbitrage activity. Thus, anomaly-based strategies tend to be more profitable when market makers face difficulties in providing liquidity. These liquidity dry-ups either occur during times of financial market turmoil ([Nagel, 2012](#)), or when the capital ratio of financial intermediaries is low. Finally, investor sentiment may also impact the profitability of anomaly strategies by creating mispricing in the presence of short sale constraints ([Stambaugh, Yu, and Yuan, 2012](#)).

I divide my sample according to six measures related to various market states. First, I use overall market return benchmarks based on value weighted stock and dollar open interest weighted option portfolios. Second, I identify periods with low market making capacity using [Pástor and Stambaugh](#)’s liquidity measure, the VIX index, and [He et al.](#)’s intermediary

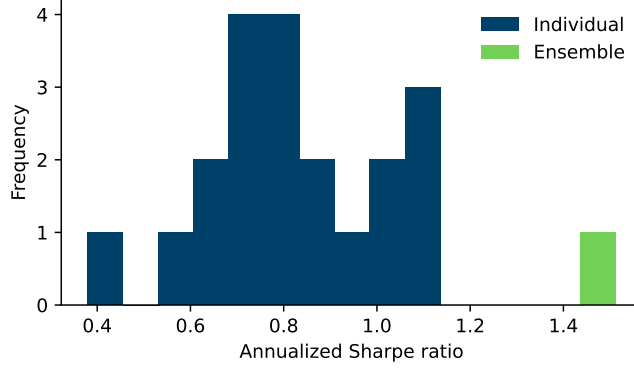


Figure 3: Annualized Sharpe ratios of long-short option portfolios. The figure shows the distribution of Sharpe ratios obtained from long-short option portfolios based on individual CNN forecasts (blue bars) and the ensemble forecast, i.e. a simple average across models (green bar). Portfolios are weighted by dollar open interest at portfolio formation.

capital ratio. Finally, high sentiment periods are measured using [Baker and Wurgler \(2007\)](#) investor sentiment.¹⁵

I define sub-periods relative to the 5th and 95th percentile of each distribution over my sample period. Table III reports average return, standard deviation, and annualized Sharpe ratio of the CNN portfolio during these extreme months. The table further reports the t-statistic of a test for the difference in means between the two samples. I find significant evidence that profitability of the CNN portfolio is affected by the risk-taking capacity of financial intermediaries. Specifically, the CNN based option strategy exhibits negative expected returns when intermediaries have an abundance of equity capital. In contrast, the annualized Sharpe ratio exceeds two when the aggregate capital ratio is low. The remaining scenarios do not exhibit significant differences in expected returns, however Sharpe ratios are notably elevated in months with high VIX or extreme stock market returns. According to theory, these periods of heightened uncertainty are associated with tighter funding constraints and lower market-making capacity, suggesting that the CNN based option portfolio is rewarded for counter-cyclical liquidity provision ([Gromb and Vayanos, 2002](#); [Brunnermeier and Pedersen, 2009](#); [Adrian and Shin, 2010](#)).

Portfolio composition. I proceed to examine the composition of portfolios formed by

¹⁵I source the value-weighted stock market return from [Kenneth French's Data Library](#). Aggregate market liquidity is from [Robert Stambaugh's website](#) and the VIX is available on the [CBOE website](#). Intermediary capital ratios are provided on [Zhiguo He's website](#) and sentiment is taken from [Jeffrey Wurgler's website](#). Note that investment periods typically end on the third Friday each month, causing them to be imperfectly aligned with measures reported at the end of the month.

	Obs.	Mean	Sd.	Sharpe	Diff.	t-statistic
<i>Stock market return</i>						
Low	12	4.19	4.99	2.90		
High	12	5.07	5.45	3.22	0.88	0.41
<i>Option market return</i>						
Low	12	3.63	7.91	1.59		
High	12	1.90	4.59	1.44	-1.73	-0.65
<i>Stock market liquidity</i>						
Low	12	3.56	5.74	2.15		
High	12	3.63	4.47	2.81	0.07	0.03
<i>VIX</i>						
Low	12	2.16	6.14	1.22		
High	12	4.20	5.49	2.65	2.04	0.86
<i>Capital ratio</i>						
Low	12	3.13	5.06	2.14		
High	12	-0.66	4.15	-0.55	-3.79	-2.00
<i>Investor sentiment</i>						
Low	12	4.42	6.58	2.33		
High	12	5.18	10.08	1.78	0.75	0.22

Table III: Performance of CNN portfolio by market state. This table shows summary statistics (number of observations, average and standard deviation of the monthly return, and Sharpe ratio) of the CNN based option portfolio during different sub-periods. I consider months during which the value-weighted stock market portfolio, dollar open interest weighted option market portfolio, [Pástor and Stambaugh](#) stock liquidity, VIX, intermediary capital ratio, and [Baker and Wurgler](#) sentiment are lower (higher) than the 5th (95th) percentile of their sample distribution. The table also reports t-statistics for the difference between the average return in the two sub-periods. Data are from 2003 to 2022.

sorting options into deciles based on CNN predictions. Table IV reports descriptive statistics for a selected set of option and stock characteristics. The results indicate that the high-minus-low (HML) portfolio exhibits economically and statistically significant exposures to Gamma and Theta.¹⁶ In contrast, exposures to Delta and Vega are economically small. I further find that options being sold by the CNN tend to have higher implied volatilities than those being bought. The table also shows average option demand pressure as measured by the total option open interest scaled by equity market capitalization following Zhan et al. (2022). Demand pressure exhibits a pronounced U-shape across CNN deciles, with values in the extreme deciles that are four to five times larger than in the center. This suggests that the CNN effectively identifies options subject to significant demand pressure. These results are consistent with prior evidence linking demand pressure to the shape of the implied volatility surface (Bollen and Whaley, 2004; Garleanu et al., 2009). Finally, I do not find significant exposures of the HML portfolio to underlying stock characteristics. Nevertheless, there are systematic patterns across the deciles. For example, relative to the middle portfolios, stocks in the extreme deciles are characterized by smaller market capitalization (around \$10 billion) and lower investment levels, while exhibiting higher market betas.

4.2. Impact of transaction costs

It is common in empirical options research to assume that options are traded at the midpoint of quoted bid and ask prices. However, since spreads in option markets can be large, I now turn to the question of whether the CNN portfolio remains profitable after incorporating option transaction costs. To do this, I follow Zhan et al. (2022) and assume that the realized effective spread is a fraction of the quoted spread. I also adjust the predicted return before sorting options into deciles. This can be achieved without making any assumptions about future spreads because positions are held until expiration. The adjusted prediction $\hat{r}_{t+1}^{dh}$ is

$$\hat{r}_{t+1}^{dh,TC} = \frac{(\hat{r}_{t+1}^{dh} + 1 + r_f)(\Delta_t \cdot S_t - C_t)}{\Delta_t \cdot S_t - (C_t \pm \text{spread}_t)} - 1 - r_f, \quad (4)$$

where spread_t is the assumed effective spread in dollars, that is added to (subtracted from) the call price in case of buying (selling) the option.

¹⁶Note that the CNN model predicts delta-neutral call writing returns, resulting in an HML portfolio which sells options in the top decile and buys options in the bottom decile.

	Option characteristics							Stock characteristics				
	Spread	Delta	Gamma	Vega	Theta	IV	ODP	Size	Beta	BM	Inv	GP/AT
Low	13.47	0.54	2.88	0.10	-0.47	0.68	25.93	14.52	1.56	0.39	-0.49	0.22
2	8.97	0.53	5.53	0.10	-0.25	0.34	7.20	91.34	0.99	0.39	-0.20	0.33
3	6.83	0.53	6.02	0.10	-0.21	0.28	4.08	151.54	0.80	0.36	-0.16	0.34
4	6.54	0.53	5.72	0.10	-0.22	0.29	4.21	145.60	0.82	0.35	-0.17	0.34
5	6.79	0.53	5.16	0.10	-0.23	0.32	4.29	114.73	0.88	0.37	-0.17	0.33
6	7.22	0.54	4.62	0.10	-0.26	0.35	5.62	80.20	0.98	0.37	-0.19	0.34
7	7.70	0.54	4.11	0.11	-0.29	0.39	6.45	59.40	1.07	0.38	-0.21	0.34
8	8.40	0.54	3.58	0.11	-0.32	0.44	8.37	46.45	1.18	0.39	-0.22	0.33
9	10.11	0.54	3.07	0.10	-0.38	0.52	10.39	24.38	1.31	0.41	-0.27	0.32
High	12.94	0.54	2.23	0.10	-0.54	0.75	17.83	9.68	1.54	0.39	-0.41	0.23
HML	-0.52	0.01	-0.66	0.00	-0.07	0.07	-8.10	-4.83	-0.03	0.00	0.08	0.01
<i>t</i>	(-0.81)	(2.76)	(-10.58)	(0.15)	(-4.39)	(3.35)	(-3.52)	(-1.77)	(-0.68)	(0.26)	(1.25)	(1.28)

Table IV: Descriptive statistics for CNN portfolios. This table shows time-series averages of dollar open interest weighted average characteristics of decile portfolios sorted by CNN model predictions. Spread is the quoted percentage bid-ask spread, Delta is the option’s delta, IV is the average implied volatility of an at-the-money call and put. Option demand pressure (ODP) is option open interest divided by stock volume times 100. Size measures a stock’s market capitalization at the end of the prior month in billions of dollars. Beta is the slope from a rolling regression of monthly excess stock returns on the market excess return in a 60-month window. Inv is the change in total assets, and GP/AT is gross profits divided by total assets. *t*-statistics adjusted for heteroscedasticity and autocorrelation are reported in parentheses. The sample covers at-the-money call options from January 2003 to December 2022.

Table V shows the performance of the CNN portfolio for varying fractions of effective spread relative to quoted bid-ask spread. Panel (a) reports results for adjusted return predictions. Profitability decreases monotonically in option transaction costs but remains positive and statistically significant at all levels. Specifically, the average return drops from 2.77% to 0.56% per month as the effective spread moves from 0% to 100% of the quoted spread. Empirical evidence from prior studies shows that actual effective bid-ask spreads of equity options are roughly 50% of the quoted spread (Mayhew, 2002; De Fontnouvelle, Fishe, and Harris, 2003; Zhan et al., 2022). In this case the long-short portfolio would earn 1.03% per month with a *t*-statistic above 4. Nevertheless, this number likely represents a lower bound on the return achievable by sophisticated investors. According to Muravyev and Pearson (2020), investors timing their orders intraday can lower the conventional half spread by 63%.

Panel (b) shows the performance of machine learning portfolios based on unadjusted return predictions. The results indicate that adjusting return predictions by current spreads only becomes relevant for relatively high effective spreads.

Eff. spread	$\mathbb{E}[r]$	t	SR
0%	2.77	(7.75)	1.47
25%	1.64	(6.11)	1.15
50%	1.03	(4.23)	0.85
75%	0.78	(3.73)	0.78
100%	0.56	(3.17)	0.62

(a) Adjusted return predictions ($\hat{r}_{t+1}^{dh,TC}$)

Eff. spread	$\mathbb{E}[r]$	t	SR
0%	2.77	(7.75)	1.47
25%	2.23	(6.36)	1.19
50%	1.69	(4.88)	0.90
75%	1.15	(3.33)	0.62
100%	0.61	(1.76)	0.33

(b) Unadjusted return predictions ($\hat{r}_{t+1}^{dh}$)

Table V: Impact of transaction costs. The table shows average returns, t-statistics, and Sharpe ratios of dollar open interest weighted long-short portfolios for varying effective spreads. Effective spreads are computed as a fraction of the quoted spread. Panel (a) shows strategies based on predictions of CNNs trained on option mid prices, whereas models in Panel (b) are trained on prices using the prevailing effective spread. t-statistics adjusted for heteroscedasticity and autocorrelation are reported in parentheses. Data are at the monthly frequency and span January 2003 to December 2022.

4.3. Retail trading and the shape of the surface

The results in Section 4.1 indicate that the CNN strategy provides liquidity in options with relatively high demand pressure, and that it is more profitable when intermediaries are constrained. This naturally raises the question of who the strategy provides liquidity to. Recent evidence shows that retail investors have become the dominant group in the options market, accounting for 62% of total trading volume (Bryzgalova et al., 2023). Retail activity has also been shown to be heavily tilted toward buy trades, outnumbering sell trades by a factor of seven, likely reflecting broker restrictions on uncovered selling (Bogousslavsky and Muravyev, 2024). Such imbalances can generate demand pressure, pushing prices upward when intermediaries cannot perfectly hedge their positions (Garleanu et al., 2009). To test whether the CNN strategy detects retail-induced price pressure, I examine whether predicted returns of delta-neutral call-writing are positively related to retail trading activity in the corresponding options.

Following Bryzgalova et al. (2023), I identify retail trades as those trades executed through a single-leg non-ISO price improvement mechanism (SLIM).¹⁷ I then measure retail activity as the share of retail trading volume in total option trading volume. Next, I regress the retail share on same-day CNN predictions and a large number of covariates listed in Appendix A.1. Table VI shows that delta-neutral call-writing profits are concentrated in options with high retail participation. Specifically, a one percentage point increase in the rank of CNN predictions is associated with a 0.6 to 2.7 percentage points higher share of retail volume.

¹⁷Data are available at www.sikorskaya.net/research.

	SLIM volume share		SLIM dollar volume share	
	(1)	(2)	(3)	(4)
$\hat{r}_{t+1}^{CNN}$	2.2*** (5.24)	0.61** (2.3)	2.68*** (4.63)	0.89* (1.85)
Controls	No	Yes	No	Yes
Observations	1746.25	1734.65	1746.25	1734.65
Adj. R^2	0.98	14.34	1.42	14.44

Table VI: Cross-sectional regressions to explain retail trading activity in options. The table shows average slopes ($\times 100$) and t-statistics (in parentheses) from [Fama and MacBeth](#) cross-sectional regressions of various measures of retail trading activity in options on CNN predictions and a set of control variables. SLIM (dollar) volume is the total (dollar) trading volume in all call options on the same underlying executed through single-leg non-ISO price improvement mechanisms ([Bryzgalova et al., 2023](#)). The volume shares are computed as a fraction of total option trading volume on that day. The table further reports the average number of observations and adjusted R^2 in percent. I estimate regressions using weighted-least squares with dollar open interest as the weight. Independent variables are first standardized by their cross-sectional rank and then scaled to the interval $[0, 1]$. t-statistics are adjusted for heteroscedasticity and autocorrelation ([Newey and West, 1987](#)). Detailed variable definitions are provided in [Appendix A.1](#). Data are at the monthly frequency from November 2019 to June 2021.

This suggests that the CNN is able to identify options with high retail activity, consistent with prior evidence that (retail) demand pressure affects the shape of the implied volatility surface ([Bollen and Whaley, 2004](#); [Eaton et al., 2024](#)).

4.4. Association with option and stock characteristics

Predictive regressions. I use cross-sectional regressions to investigate whether the CNN model remains a significant return predictor after controlling for variables known to explain the cross-section of equity options. Furthermore, the analysis allows me to assess which variables are subsumed by the CNN. I consider option predictors that can be categorized into (i) risk-neutral measures, (ii) variables describing the implied volatility surface, (iii) variables combining information from stock and option markets, and, (iv) underlying stock fundamentals. [Appendix A.1](#) details the construction of these variables.

[Table VII](#) reports average slopes from [Fama and MacBeth](#) cross-sectional regressions. Each month, I regress individual option returns on lagged predictors and CNN model forecasts. I use weighted least squares regressions with dollar open interest as weight. Independent variables are rank-standardized from 0 to 1 to ease interpretation. The first regression (column 1) shows that among several risk-neutral measures, only implied variance and its term structure are associated with future option returns. Both variables lose their statistical

significance after introducing CNN predictions (column 2), indicating that the model has “learned” these patterns. Among the second set of variables (columns 3 and 4), the spread between at-the-money call and put implied volatility (Bali and Hovakimian, 2009; Cremers and Weinbaum, 2010) as well as the slope of the implied volatility term structure (Vasquez, 2017) are highly significant predictors. Including the CNN model reduces, but not diminishes the economic and statistical significance of both predictors. The final set of variables combines option information with information from the underlying stock, which is unobservable to the CNN (columns 5 and 6). I find that option demand pressure, realized variance (RV), as well as the variance risk premium (VRP) are significantly related to individual option returns. Perhaps surprisingly, the predictive ability of RV and VRP vanishes after controlling for CNN forecasts. The last column demonstrates that the CNN robustly predicts future option returns even after adding a host of control variables. Furthermore, only two predictors, namely the call-put spread and option demand pressure (ODP), remain highly significant.

Prior studies have introduced multiple approaches to measure the variance risk premium (VRP). Differences commonly arise from using model-free implied variance versus the average of call and put implied volatilities derived from a Cox, Ross, and Rubinstein binomial tree model (e.g., Bakshi and Kapadia, 2003; Bali and Hovakimian, 2009; Carr and Wu, 2009; Goyal and Saretto, 2009), as well as from the choice of historical window for realized variance. Table A.1 reports cross-sectional regressions of future option returns on CNN predictions and four alternative VRP measures. I find that measures with longer look-back windows are the strongest predictors and retain significance when CNN predictions are included. Nevertheless, none of the VRP measures subsume the CNN model, and their predictive power disappears when all measures are included jointly.

Table A.2 examines the predictive power of selected underlying stock characteristics from Chen and Zimmermann (2022). Prior work documents that variables such as profitability, cash holdings, cash flow variance, analyst forecast dispersion, and new share issuance are strongly correlated with future option returns (Zhan et al., 2022). Consistent with Duarte et al. (2025), I do not find similarly strong effects in my option sample, which is carefully constructed to avoid look-ahead bias. In contrast, CNN predictions remain robustly associated with future option returns and are not explained by these underlying stock characteristics. These results suggest that the CNN model captures patterns that are distinct from previously

documented option return predictors.

Contemporaneous regressions. To further tease out the association with commonly studied predictors, I estimate period-by-period cross-sectional regressions of CNN predictions on contemporaneous option-based variables. For ease of interpretation, I use cross-sectional ranks of each variable, which are then scaled to the interval $[0, 1]$. Table A.3 tabulates average slopes, t-statistics (in parentheses), and adjusted R-squared values. The results show that CNN predictions load significantly on known option characteristics, such as the call-put spread, risk-neutral downside variance, or the slope of the volatility term structure. However, only less than 50% of the cross-sectional variation in CNN predictions can be explained by the large set of commonly studied option characteristics.

4.5. Factor model alphas

The literature on factor models for equity options is relatively sparse, however recent studies have succeeded in explaining a significant portion of option returns using Instrumented Principal Component Analysis (IPCA, Kelly, Pruitt, and Su, 2020) (Büchner and Kelly, 2022; Goyal and Saretto, 2025).¹⁸ IPCA uses a potentially large set of observable characteristics to instrument for dynamic factor loadings. Characteristics map linearly to betas and the corresponding factors are latent. I estimate IPCA using an expanding window identical to the one used to train the CNN, but roll it forward after each month instead of just once per year. I consider three sets of observable asset characteristics as instruments. The first contains only underlying stock characteristics from Chen and Zimmermann (2022), the second includes option characteristics studied in Goyal and Saretto (2025); Neuhierl, Tang, Varneskov, and Zhou (2025), and the third set covers both. Details on model estimation and evaluation can be found in Appendix A.2.

Table A.4 in the Internet Appendix demonstrates the model’s ability to accurately describe option returns. The total R^2 generally increases in the number of latent factors and ranges from 9% to 12%.¹⁹ The total R^2 is relatively similar across different characteristics sets, but this is not the case for the predictive R^2 . For example, the predictive R^2 of IPCA with five latent factors amounts to 0.14% using only stock characteristics, and more than

¹⁸IPCA has also been shown to work well empirically in other asset classes, such as stocks (Kelly et al., 2019), corporate bonds (Kelly, Palhares, and Pruitt, 2023), and cryptocurrencies (Bianchi and Babiak, 2021).

¹⁹Goyal and Saretto (2025) report a total R^2 of 9.61% for a similar sample.

dep = r_{t+1}^{dh}								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{r}_{t+1}^{CNN}$		2.13*** (6.14)		1.75*** (4.48)		2.48*** (6.33)		1.70*** (4.33)
Var ^Q	5.56** (2.04)	3.48 (1.22)					3.49 (1.13)	2.65 (0.84)
Skew ^Q	-0.23 (-0.32)	-0.21 (-0.29)					-0.06 (-0.11)	-0.11 (-0.19)
Kurt ^Q	0.30 (0.36)	0.22 (0.27)					0.97 (1.19)	0.52 (0.68)
DownVar ^Q	-5.18* (-1.76)	-4.59 (-1.53)					-1.51 (-0.47)	-1.29 (-0.39)
TsVar ^Q	1.11*** (2.89)	0.51 (1.21)					0.23 (0.47)	0.22 (0.46)
TsSkew ^Q	0.27 (0.46)	0.05 (0.09)					-0.10 (-0.18)	-0.08 (-0.14)
TsKurt ^Q	-0.27 (-0.34)	0.01 (0.02)					-0.49 (-0.67)	-0.19 (-0.26)
IVolATM			0.31 (0.99)	-0.79*** (-2.44)			-0.24 (-0.22)	-0.69 (-0.64)
C-P Spread			1.02*** (4.25)	0.70*** (2.66)			0.79*** (3.52)	0.57*** (2.38)
IVSlope			-1.15*** (-3.87)	-0.69** (-2.23)			-0.96** (-2.25)	-0.50 (-1.23)
IVSkew			-0.38 (-1.23)	-0.51* (-1.67)			-0.98** (-2.04)	-0.92* (-1.92)
VoV			0.38* (1.80)	0.38* (1.83)			0.33 (1.57)	0.37* (1.82)
OS					0.09 (0.34)	0.22 (0.81)	-0.06 (-0.25)	0.01 (0.05)
ODP					-1.10*** (-3.57)	-1.17*** (-3.88)	-0.96*** (-3.26)	-1.07*** (-3.56)
Var ^P					1.14*** (3.71)	-0.16 (-0.53)	-0.20 (-0.42)	-0.37 (-0.77)
VRP					0.65** (2.13)	-0.53 (-1.56)	-0.57 (-1.16)	-0.65 (-1.31)
Adj. R^2	7.2	7.86	5.89	6.63	4.49	5.67	12.16	12.74

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table VII: Cross-sectional regressions to predict option returns. The table shows average slopes ($\times 100$) and t-statistics (in parentheses) from Fama and MacBeth cross-sectional regressions of delta-hedged option returns on CNN predictions and a set of control variables. The table further reports the average adjusted R^2 in percent. I estimate regressions using weighted-least squares with dollar open interest as the weight. Independent variables are first standardized by their cross-sectional rank and then scaled to the interval $[0, 1]$. t-statistics are adjusted for heteroscedasticity and autocorrelation (Newey and West, 1987). Detailed variable definitions are provided in Appendix A.1. Data are at the monthly frequency from January 2003 to December 2022.

	Number of latent factors					
	1	2	3	4	5	6
$\alpha_{\text{Stock}}^{\text{IPCA}}$	2.75 (7.59)	2.62 (7.85)	2.75 (6.75)	2.33 (6.06)	2.39 (5.52)	2.19 (4.62)
$\alpha_{\text{Option}}^{\text{IPCA}}$	2.78 (7.69)	1.87 (3.96)	1.77 (3.17)	1.38 (2.35)	1.65 (3.15)	1.65 (3.85)
$\alpha_{\text{Full}}^{\text{IPCA}}$	2.75 (7.56)	1.56 (3.17)	2.1 (4.88)	1.76 (4.02)	1.73 (3.06)	1.64 (3.18)

Table VIII: IPCA alphas. The table shows intercepts ($\times 100$) and t-statistics (in parentheses) from time-series regressions of the long-short CNN portfolio on IPCA factors. I estimate IPCA using underlying stock characteristics, option characteristics, or both. t-statistics are adjusted for heteroscedasticity and autocorrelation (Newey and West, 1987). Data are at the monthly frequency from January 2003 to December 2022.

doubles to 0.38% when using option characteristics. This is also reflected in the economic performance of out-of-sample long-short portfolios. These are constructed by sorting options into decile portfolios based on their conditional expected return. The resulting Sharpe ratio of a dollar open interest weighted (OIW) portfolio is below 0.6 using only stock characteristics, but generally exceeds 1 when option characteristics are added.

I benchmark the CNN against latent factors obtained from various IPCA specifications. Table VIII reports monthly alphas from time series regressions of the CNN portfolio on factor returns. I consider 1 to 6 latent factors and different characteristics sets. The striking result is that the CNN portfolio generates economically and statistically significant alphas across all specifications. For example, with the common choice of $K = 5$, the alpha amounts to 2.39% per month with a t-statistic above 5 when measured against IPCA with stock characteristics from Chen and Zimmermann (2022). Adding option characteristics reduces the alpha to 1.73% per month (t-statistic of 3.06), which is still sizable. In summary, I find that exposure to systematic risk factors in the option market cannot account for the high returns associated with the CNN based machine learning portfolio.

4.6. Model comparison

I now proceed to benchmark the CNN performance against other machine learning portfolios. For this exercise, I consider the IPCA model based on pre-specified option and stock characteristics, and a fully connected neural network (FCNN). The FCNN receives the flattened volatility surface as input and does not explicitly account for the grid-like topology of the

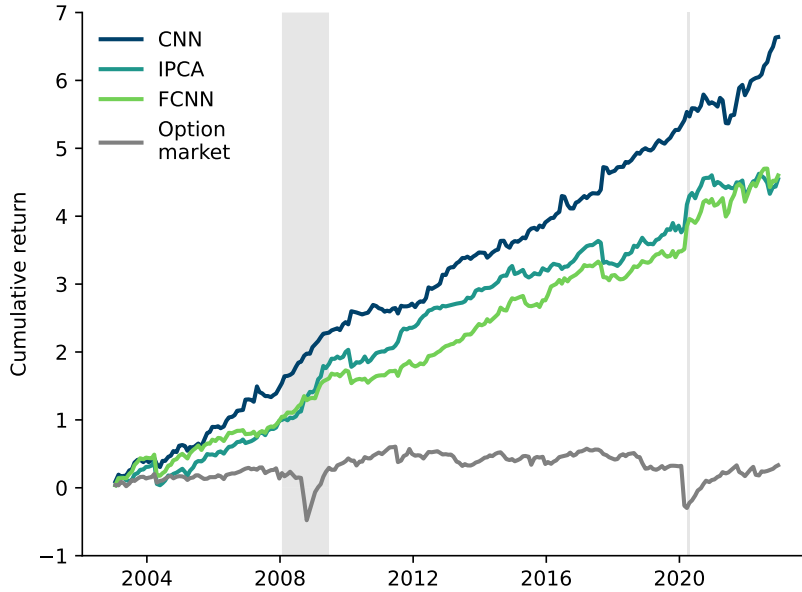


Figure 4: Cumulative excess returns of machine learning portfolios. The figure displays cumulative excess returns of machine learning portfolios. Machine learning portfolios are formed by sorting options into deciles based on their conditional expected return. The grey line represents cumulative excess returns of a portfolio of all delta-neutral calls included in the sample. All portfolios are weighted by dollar open interest at the beginning of the investment period and adjusted to match volatility of the CNN portfolio. Shaded areas indicate NBER recession periods. The monthly sample is from 2002 to 2022.

data.²⁰

Time-series analysis. As before, I construct machine learning portfolios by sorting options into deciles based on their out-of-sample conditional expected return. Figure 4 shows cumulative excess returns of dollar open interest weighted decile spreads and an overall option market portfolio. The benchmarks are volatility adjusted relative to the CNN portfolio. All machine learning strategies beat the market by a wide margin and, unlike the market, did not exhibit pronounced drawdowns during the global financial crisis and the COVID-19 shock. Furthermore, the CNN portfolio consistently outperforms the FCNN and IPCA portfolio. I run time series regressions to investigate whether the CNN generates alpha relative to competing models. The results in Table IX show that none of the benchmarks have explanatory power for the CNN portfolio. Even including all benchmarks simultaneously yields a large and statistically significant alpha of 2.4% per month.

²⁰The FCNN architecture closely resembles the CNN, but uses fully connected layers instead of convolutional layers. The number of neurons in each layer is 128, 64, 32, 16 and the network is regularized, optimized, and trained as described in Section 2.

	dep = CNN						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
const	2.77*** (7.75)	2.78*** (7.74)	2.37*** (4.56)	2.77*** (6.74)	2.38*** (3.94)	2.65*** (5.25)	2.41*** (4.61)
Option market		-0.08 (-0.77)					-0.21*** (-2.39)
FCNN			0.21 (1.61)				0.44*** (2.49)
IPCA (stock)				-0.01 (-0.07)			-0.21 (-1.64)
IPCA (option)					0.22 (1.13)		0.07 (0.32)
IPCA (full)						0.06 (0.43)	-0.20 (-1.25)
Observations	240	240	240	240	240	240	240
Adj. R^2	0.0	-0.28	5.43	-0.41	4.52	-0.06	11.67

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table IX: Time series regressions. The table shows intercepts ($\times 100$) and t-statistics (in parentheses) from time-series regressions of the long-short CNN portfolio on the option market portfolio, FCNN portfolio, and IPCA portfolio. The option market portfolio is the average return from writing all delta-neutral call options in the sample. Long-short machine learning portfolios are formed by sorting all options into deciles based on model predictions. Portfolios are weighted by dollar open interest at portfolio formation. Data are at the monthly frequency from January 2003 to December 2022.

Cross-sectional analysis. I also run cross-sectional regressions to examine the models' predictive ability for individual equity option returns. Table X reports average slopes and t-statistics from Fama and MacBeth cross-sectional regressions using dollar open interest as weight. The first column shows that the CNN alone explains, on average, 5.6% of the cross-sectional variation in option returns. Further, a one percentage point increase in CNN prediction is associated with a one percentage point higher option return with a t-statistic of 3.6. Columns (2) to (5) show that the FCNN and IPCA models are also significant predictors of future option returns, but with lower R^2 values. The final specification (column 6) evaluates the joint explanatory power of all models. Overall, the models manage to explain more than 11% of the cross-sectional variation in individual option returns, adding significantly to the specification using only one model. The CNN remains a strong and significant predictor of future option returns, but does not fully subsume the FCNN and IPCA model.

	(1)	(2)	(3)	(4)	(5)	(6)
$\hat{r}_{t+1}^{\text{CNN}}$	0.98*** (3.64)					0.63*** (2.33)
$\hat{r}_{t+1}^{\text{FCNN}}$		0.28*** (3.67)				0.15** (2.12)
$\hat{r}_{t+1}^{\text{IPCA (stock)}}$			0.38*** (2.68)			0.10 (0.75)
$\hat{r}_{t+1}^{\text{IPCA (option)}}$				0.55*** (5.37)		0.14 (1.47)
$\hat{r}_{t+1}^{\text{IPCA (full)}}$					0.44*** (5.10)	0.28*** (3.20)
Adj. R^2	5.59	4.37	1.95	2.12	2.27	10.91

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table X: Cross-sectional regressions to predict option returns. The table shows average slopes ($\times 100$) and t-statistics (in parentheses) from Fama and MacBeth cross-sectional regressions of delta-hedged option returns on CNN, FCNN, and IPCA predictions. I estimate regressions using weighted-least squares with dollar open interest as the weight. Independent variables are first standardized by their cross-sectional rank and then scaled to the interval $[0, 1]$. t-statistics are adjusted for heteroscedasticity and autocorrelation (Newey and West, 1987). Data are at the monthly frequency from January 2003 to December 2022.

4.7. Predictor importance

I now aim to shed light on which parts of the volatility surface deem important for predicting equity option returns. This task naturally poses a challenge due to the extensive number of parameters and nonlinear associations of deep learning models. A commonly used method to explain fitted prediction models draws on insights from game theory to assign importance scores to each feature for a specific prediction (Lundberg and Lee, 2017). Specifically, I compute Shapley (SHapley Additive exPlanations, Shapley, 1953) values, which represent the average marginal contribution of a predictor relative to all possible coalitions excluding that predictor.²¹

I begin by visualizing the importance of individual points on the implied volatility surface. First, I use 1,000 random instances of my training data (*“background data”*) to integrate out predictors. Second, separately for each model and out-of-sample year, I compute Shapley values for 1,000 observations. Last, I average across all models in the ensemble. Figure 5 displays mean absolute Shapley values, where larger numbers indicate greater influence on the

²¹Calculating Shapley values requires exponential time complexity, making it extremely expensive to compute exact values for a large number of features. To address this issue, I use the Gradient Explainer implemented in the SHAP package in python (<https://shap.readthedocs.io>). The Gradient Explainer approximates Shapley values using a sampling-based approach that leverages the gradient of the model.

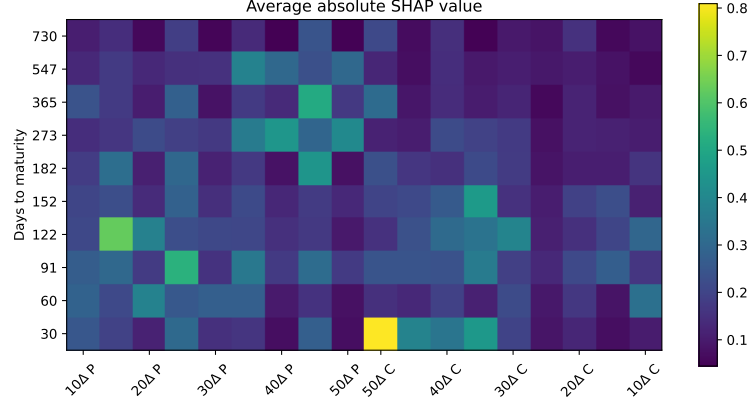


Figure 5: Shapley values. The figure shows the importance of individual implied volatility values, measured following [Lundberg and Lee \(2017\)](#). For each model in each year, I use 1,000 random “background” images from the training data to integrate out predictors. Then, I compute Shapley values for 1,000 randomly drawn out-of-sample observations. Due to the linearity property of Shapley values, I obtain ensemble Shapley values by averaging across all models. Finally, feature importance is computed as the average absolute Shapley value of all out-of-sample observations. Data is from 1996 to 2022.

prediction outcome. The most influential point is the implied volatility of at-the-money call options with one month to expiry. This result is consistent with recent empirical evidence showing that retail investors strongly prefer calls over puts, and are most active near the strike price ([Bryzgalova et al., 2023](#); [Bogousslavsky and Muravyev, 2024](#); [Eaton et al., 2024](#)). Other important predictors cluster around short-term, deep out-of-the-money put options. This finding might be related to the preference of retail investors for lottery-like payoffs ([Boyer and Vorkink, 2014](#); [Filippou et al., 2018](#)). In contrast, parts of the surface corresponding to longer maturities, particularly out-of-the-money calls, are relatively unimportant. Taken together, these patterns indicate that the CNN extracts information from regions of the surface where retail activity is empirically most pronounced.

I also form feature groups to obtain a clear-cut picture of importances across different dimensions of the volatility surface. Figure 6 aggregates Shapley values of individual points on the implied volatility surface. Panel (a) analyzes different delta groups and displays the dispersion of values in the years of the testing period. The figure shows that implied volatility of deep out-of-the-money calls is much less important than of at-the-money options or deep out-of-the-money puts. Panel (b) reports aggregate Shapley values across maturities. Importances tend to decrease in time to maturity. On average, information from options with approximately one quarter until expiration is most important for predicting future option

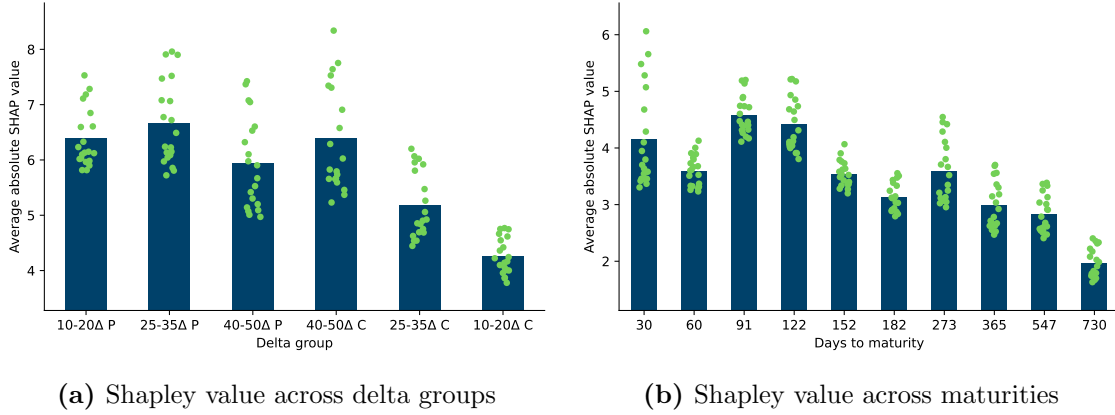


Figure 6: Feature group importances. The figure shows the importance of feature groups, measured by Shapley values following [Lundberg and Lee \(2017\)](#). The group importance is the sum of individual Shapley values across all features in the respective group. Each dot represents an individual year in the testing sample, whereas bars are computed over the entire testing sample.

returns. However, the importance of one-month implied volatility shows a high dispersion, being the most important predictor group during four years of the testing period.

4.8. Visual analysis of CNN predictions

This section offers an additional perspective on the interpretation of CNN model predictions by visualizing the volatility surfaces associated with low and high forecasted returns. Following [Murray et al. \(2024\)](#), I compute dollar open interest weighted average volatility surfaces for each decile portfolio in each month. Figure 7 plots the time series averages of these surfaces for the bottom and top decile portfolios.²² My results reveal differences between extreme decile portfolios that are easily discernible to the human eye. First, the bottom decile exhibits a pronounced spread at-the-money that is largely absent in the top decile. Second, the term structure of implied volatility is slightly upward-sloping for the bottom decile but distinctly downward-sloping for the top decile. Third, options in the bottom decile display a steeper smirk compared with the top decile. Overall, these visual patterns align with regions of high Shapley values, providing independent validation of the model’s feature importance results (see Section 4.7). The observed differences are also in line with cross-sectional regression results from Section 4.4 and with prior literature on the call-put spread ([Bali and Hovakimian, 2009](#); [Cremers and Weinbaum, 2010](#)), risk-neutral skewness ([Bali and Murray, 2013](#)), and the slope of the volatility term structure ([Vasquez, 2017](#)).

²²Results for all remaining deciles are displayed in Figure A.5 in the Internet Appendix.

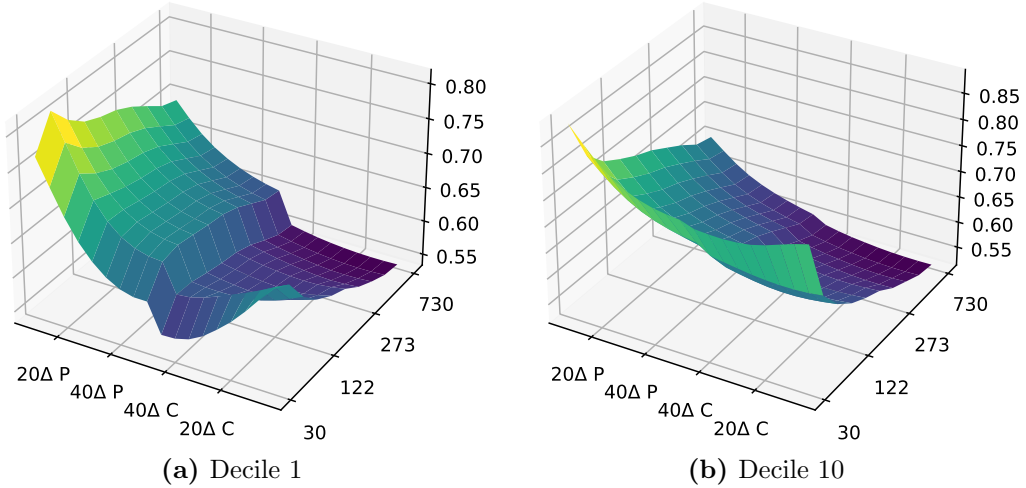


Figure 7: Average volatility surfaces of CNN decile portfolios. This figure shows the average volatility surface for options within the bottom and top CNN decile portfolio. First, at each portfolio formation date and separately for each decile portfolio, I calculate the dollar open interest weighted average of each individual value on the volatility surface. I then take the time-series averages of these monthly means for each decile portfolio and implied volatility. The sample period is monthly and spans January 2003 to December 2022.

5. ROBUSTNESS

This section provides additional results and robustness checks. I examine a higher hedging frequency (Section 5.1) and provide results for alternative option positions (Section 5.2). Finally, Section 5.3 demonstrates the benefits of frequent model fine-tuning.

5.1. Hedging frequency

My previous results are based on call options that are hedged once with delta shares of the underlying stock at the time of trade initiation. The static hedge, however, carries the risk of losing delta neutrality due to spot movements over the life of the option. To alleviate concerns that the CNN model predicts directional risks from the underlying, I proceed to examine returns of option positions with a daily rehedging schedule. According to [Tian and Wu \(2023\)](#), daily delta hedging reduces directional risk by more than 90% relative to a naked option position, compared to 70% for the static hedge at initiation.

Following [Bakshi and Kapadia \(2003\)](#), [Cao and Han \(2013\)](#), and [Büchner and Kelly](#)

(2022), the profit and loss of buying an option that is delta-hedged daily is given by

$$\begin{aligned} \Pi(t, t + \tau) = & \max \{S_{t+\tau} - K, 0\} - V_t - \sum_{n=0}^{N-1} \Delta_{t_n} \times [S_{t_{n+1}} - S_{t_n}] \\ & - \sum_{n=0}^{N-1} r_{t_n} \times \frac{a_{t_n}}{365} \times [V_{t_n} - \Delta_{t_n} S_{t_n}], \end{aligned} \quad (5)$$

where V_t is the option mid price and S_t the closing stock price on day t . Δ_t denotes the option delta on day t , r_t the annualized risk-free rate on day t , and a_t the number of calendar days between t and $t + 1$. Delta-hedged option returns are given by the profit and loss, scaled by the absolute price of the initial portfolio:

$$r_{t,t+\tau} = \frac{\Pi(t, t + \tau)}{|\Delta_t S_t - V_t|}. \quad (6)$$

I use end-of-day closing prices to compute the profit and loss following Equation 5. The results are summarized in Figure A.3. Panel (a) shows the distribution of delta-hedged option returns with static and daily hedging frequency. Visually, the distribution of returns with daily rebalancing appears more leptokurtic, but is otherwise relatively similar to the distribution of returns with a static hedge. I further note that the number of reheding days differs from the total number of days in the investment month due to data availability. Panel (b) displays the cumulative distribution of the number of available reheding days within each investment month. The graph indicates that more than 90% of options included in the sample can be hedged on at least 15 days.

I now sort options into the same deciles as before, i.e. without retraining the CNN model. I then track the returns of a decile spread portfolio that hedges option positions daily.²³ For comparison, I also show the performance of a long-short portfolio in the underlying stock. Table XI presents average monthly returns, t-statistics, and Sharpe ratios of dollar open interest weighted portfolios. The results show that option portfolios with higher hedging frequency are also profitable, earning an average return of 1.4% per month (t-statistic of 4.2). In contrast, CNN predictions of delta-neutral call returns are not significantly associated with underlying stock returns. Taken together, these results suggest that the CNN model

²³I only keep option-month observations for which the delta hedge can be rebalanced at least 15 times.

	Delta-hedged option		Underlying
	Static hedge	Daily hedge	
$\mathbb{E}[r]$	2.39	1.37	0.35
t-value	(7.41)	(4.19)	(0.8)
Sharpe	1.61	0.96	0.16

Table XI: CNN portfolios. This table shows average excess returns, t-statistics, and Sharpe ratios of decile portfolios based on predictions of the return to delta-neutral call writing. Portfolios are weighted by dollar open interest at portfolio formation. I remove option-month observations with less than 15 available re-hedging days. The monthly sample spans 2003 to 2022.

successfully identifies investment opportunities in the option rather than the stock market.

5.2. Alternative option positions

I now move on to explore whether CNN predictions can be exploited using alternative option positions. This not only enhances the understanding of the CNN model, but is also relevant in the case of short selling constraints in the underlying stock. Furthermore, institutional investors may face investment mandates that restrict them to long option positions. For these reasons, I explore three alternative strategies. First, I construct long-short portfolios of delta-neutral puts, which should theoretically have the same risk profile as corresponding delta-neutral calls due to put-call parity. Second, I study long-short straddle portfolios, which represent another common instrument for trading volatility (e.g. in [Heston et al., 2023](#)). Third, I examine a “long only” portfolio that buys delta-hedged calls (puts) when the predicted return of writing delta-neutral calls is low (high).

The results are summarized in Table [XII](#). Focusing on option-month observations in which both call and put are available, slightly reduces profits of delta-neutral call portfolios from 2.8% per month to 2.4% per month. However, sorting puts into portfolios based on the same predictions more than halves the average return to 0.9% per month.²⁴ This finding suggests that delta-neutral puts have different risk-return profiles from delta-neutral calls, potentially due to demand induced price pressure ([Ramachandran and Tayal, 2021](#)). Next, I compute delta-neutral straddles from at-the-money call/put pairs following (e.g. as in [Heston et al., 2023](#)). Specifically, the straddle return is a weighted average of naked option returns, i.e. $r^{Straddle} = \omega \times r^C + (1 - \omega) \times r^P$, where the weights are chosen in proportion to the options’ mid price and delta: $\omega = \frac{-\Delta^P C}{-\Delta^P C + \Delta^C P}$. My results show that straddles portfolios based on

²⁴Delta-hedged put returns are computed as $r_{t+1}^{dh} = \frac{\Delta_t \cdot S_{t+1} - \max(K - S_{t+1}, 0)}{\Delta_t \cdot S_t - P_t} - 1 - r_t^f$.

	Calls	Puts	Straddles	Long only
$E[r]$	2.41	0.92	10.97	2.38
t-value	(6.79)	(3.29)	(5.49)	(8.48)
Sharpe	1.39	0.71	1.16	1.59

Table XII: Alternative CNN strategies. This table shows the average monthly return (in percent) and annualized Sharpe ratio of long/short portfolios based on CNN model predictions of delta-neutral call writing. Decile portfolios are weighted by dollar open interest at portfolio formation and held until the next major monthly option expiry. The strategy in column “Call” (“Put”) buys and sells at-the-money delta-hedged calls (puts). The straddle strategy does not involve trading in the underlying stock and achieves delta-neutrality by simultaneously buying and selling at-the-money straddles. The final column reflects a “long-only” strategy involving long positions in delta-hedged calls and puts. I remove underlyings for which I can only observe an at-the-money call option and not an at-the-money put option. The monthly sample is from January 2003 to December 2022.

CNN deciles exhibit a large and statistically significant return spread, and a corresponding Sharpe ratio of above 1. Finally, I examine a portfolio of delta-hedged options consisting of long call and long put positions. This portfolio earns average returns similar to the long-short call option strategy, but with an even higher Sharpe ratio of 1.6 per year.

To summarize, machine learning portfolios containing alternative instruments to exploit CNN predictions earn economically and statistically significant profits. More broadly, this finding suggests that the CNN identifies profitable opportunities for volatility strategies in general.

5.3. Model re-training frequency

My results so far have focused on predictions of frequently re-trained CNN models. Because training an ensemble of models is computationally costly, I proceed to show results for a static model. Specifically, rather than fine-tuning the CNN at the end of each year, I only train it once using all available data until December 2002. With this model I generate predictions for the entire out-of-sample period and form long-short decile portfolios as before. Table XIII reports average excess returns of the fine-tuned CNN (Panel a) and the static model (Panel b). Contrasting both long-short portfolios shows that average realized returns and Sharpe ratios are almost halved in the static case. This highlights the benefits of frequent retraining, either due to time-varying predictors or the incorporation of larger amounts of data.

	Pred.	Real.	t	SR		Pred.	Real.	t	SR
Low	-3.22	-1.04	(-3.03)	-0.56	Low	-3.40	-0.40	(-1.14)	-0.27
2	-0.24	-0.02	(-0.08)	-0.02	2	-1.08	-0.07	(-0.25)	-0.06
3	-0.03	-0.18	(-0.08)	-0.20	3	-0.64	-0.16	(-0.71)	-0.17
4	0.07	-0.00	(-0.01)	-0.00	4	-0.40	-0.08	(-0.39)	-0.09
5	0.16	-0.02	(-0.10)	-0.02	5	-0.19	-0.05	(-0.28)	-0.06
6	0.26	0.01	(0.03)	0.01	6	0.01	0.05	(0.21)	0.05
7	0.40	0.12	(0.47)	0.11	7	0.25	0.10	(0.49)	0.11
8	0.59	0.47	(1.76)	0.39	8	0.58	0.35	(1.59)	0.31
9	0.95	0.53	(2.08)	0.39	9	1.16	0.43	(1.71)	0.34
High	2.67	1.72	(4.92)	0.94	High	4.07	1.16	(3.01)	0.59
HML	5.89	2.77	(7.75)	1.47	HML	7.47	1.56	(3.24)	0.76

(a) Annual fine-tuning

(b) Static model

Table XIII: CNN performance depending on re-training frequency. This table shows average excess returns of decile portfolios based on out-of-sample predictions. Portfolios are dollar open interest weighted. The model in Panel (a) is fine-tuned annually, whereas the model in Panel (b) is trained only once using data from 1996 to 2002. t -statistics adjusted for heteroscedasticity and autocorrelation are reported in parentheses. Data are at the monthly frequency and span January 2003 to December 2022.

6. CONCLUSION

Understanding individual equity option markets has become increasingly important as trading volumes have surged in recent years. This paper shows that a CNN can extract information from the shape of the volatility surface to forecast delta-hedged option returns. Specifically, long-short portfolios formed on out-of-sample forecasts earn positive returns that persist after accounting for transaction costs. Further results highlight the underlying economic mechanism. Specifically, profits are concentrated during periods of tight intermediary capital, and predictive signals are strongest in contracts favored by retail investors, consistent with compensation for liquidity provision. I rule out alternative explanations by showing that the CNN’s forecasts are not subsumed by standard option characteristics, firm-level predictors, or systematic risk exposures. Feature importance measures highlight that the model emphasizes regions of the surface where retail demand is most pronounced, such as short-dated at-the-money calls and lottery-like out-of-the-money puts. Overall, the findings suggest that retail demand leaves identifiable footprints in the volatility surface that can be systematically exploited.

The results contribute to the literature on option return predictability in two ways. First, they establish the volatility surface as a rich source of information about expected returns,

complementing existing work that focuses on individual characteristics. Second, they illustrate how modern machine learning techniques can extract complex nonlinear signals from high-dimensional option data that are inaccessible to linear models. More broadly, the evidence supports the view that option markets are driven by investor demand and intermediary frictions, generating compensation for bearing demand-induced inventory risks.

Future research could extend these findings in several directions. One promising avenue is to examine whether volatility surface-based signals are linked to related asset classes, thereby shedding light on how demand effects transmit across markets. Another natural extension is to augment the CNN with a fully connected network that incorporates option contract attributes and firm-level characteristics. Such an approach would broaden the scope of the analysis to options across the entire moneyness domain. Finally, the framework developed in this paper could be applied to other structured financial data, including the term structure of bond yields or the volatility surface of index options.

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APPENDIX

A.1. OPTION RETURN PREDICTORS

I collect a large list of potential equity option return predictors which are also studied in [Goyal and Saretto \(2025\)](#). Predictors can be broadly categorized into (i) risk-neutral estimates, (ii) variables describing the shape of the implied volatility surface, (iii) measures combining option and stock market information, and, (iv) option contract characteristics.

(i) Risk-neutral measures

1. $\text{Var}^{\mathbb{Q}}$, $\text{Skew}^{\mathbb{Q}}$, $\text{Kurt}^{\mathbb{Q}}$: Risk-neutral moments ([Bakshi et al., 2003](#)), i.e. variance, skewness, and kurtosis of the risk-neutral return distribution obtained from options with 30 days to expiration. I follow the implementation provided by Grigory Vilkov (<https://doi.org/10.17605/OSF.IO/Z2486>) using the standardized volatility surface from OptionMetrics.
2. $\text{UpVar}^{\mathbb{Q}}$, $\text{DownVar}^{\mathbb{Q}}$: The upside (downside) risk-neutral variance, or positive (negative) semi-variance, is extracted from options with 30 days to expiration: $\text{Var}^{\mathbb{Q}} = \text{UpVar}^{\mathbb{Q}} + \text{DownVar}^{\mathbb{Q}}$.
3. $\text{TsVar}^{\mathbb{Q}}$, $\text{TsSkew}^{\mathbb{Q}}$, $\text{TsKurt}^{\mathbb{Q}}$: I describe the risk-neutral term structure using the difference between one-month and one-year risk-neutral moments.

(ii) Shape of the volatility surface

1. IVolATM : The average implied volatility of at-the-money call and put options with 30 days to expiration.
2. C-P Spread : The call minus put implied volatility spread ([Bali and Hovakimian, 2009](#); [Cremers and Weinbaum, 2010](#)) is the difference between implied volatilities of at-the-money call and put options with 30 days to maturity.
3. IVSlope : I compute the slope of the implied volatility term structure ([Vasquez, 2017](#)) as the difference between implied volatility of options with 365 and 30 days to maturity, i.e. $\text{IVSlope} = \text{IVolATM}_{365} - \text{IVolATM}_{30}$.
4. IVSkew : Implied volatility skew ([Xing, Zhang, and Zhao, 2010](#)) is the difference between the implied volatility of an out-of-the money put (delta of -0.20) and an at-the-money call (delta of 0.50), each with 30 days to expiration.

5. **VoV**: Volatility of volatility is computed following [Baltussen, Van Bakkum, and Van Der Grient \(2018\)](#). First, separately for calls and puts with absolute deltas of 0.50 and 30 days to maturity, I estimate mean and standard deviation of implied volatility in the 20 days before portfolio formation. I require at least 15 nonmissing values in each 20 day window. Then, I scale each standard deviation with the corresponding mean to obtain VoV_{call} and VoV_{put} . Finally, I compute volatility of volatility as $\text{VoV} = (\text{VoV}_{\text{call}} + \text{VoV}_{\text{put}}) / 2$.

(iii) Stock and option market information

1. **OS**: Option-to-stock volume ratio ([Johnson and So, 2012](#)). Using options with 5 to 30 days to expiration, I compute the OS ratio as monthly option volume divided by stock volume.
2. **ODP**: Option demand pressure ([Garleanu et al., 2009](#)). Following [Zhan et al. \(2022\)](#), I compute ODP as the total dollar open interest of all options on a stock, divided by underlying stock market capitalization.
3. **Var^P**: Realized variance. Following the literature (e.g., [Bakshi and Kapadia, 2003](#); [Carr and Wu, 2009](#)), I compute realized variance as the sum of squared log returns in the 20 days before portfolio formation.
4. **VRP**: Variance risk premium ([Bakshi and Kapadia, 2003](#); [Carr and Wu, 2009](#)), defined as the difference between risk-neutral and realized variance, i.e. $\text{VRP} = \text{Var}^{\mathbb{Q}} - \text{Var}^{\mathbb{P}}$.

(iv) Option contract characteristics

1. **mid**: Option mid price. The mid price is calculated as the average of bid and ask quotes.
2. **baspread**: Option bid-ask spread, calculated as the quoted spread divided by the mid price.
3. **doi**: Dollar open interest (open interest times mid price).
4. **volume**: Option volume at portfolio formation.
5. **delta**: Option delta as provided by OptionMetrics.

A.2. IPCA ESTIMATION

This appendix gives a brief summary of the IPCA model and refers to [Kelly et al. \(2019\)](#) for an in-depth description. IPCA is a conditional factor model with time-varying loadings that depend

on observable characteristics. In its most general form, IPCA follows the common factor structure

$$r_{i,t} = \beta_{i,t-1} f_t + \varepsilon_{i,t}, \quad (\text{A.1})$$

where f_t are latent factor realizations and $\beta_{i,t-1}$ are time-varying factor loadings. The main assumption of IPCA is that factor loadings are linear in observable characteristics. As a result, betas can be modeled as follows

$$\beta_{i,t-1} = z'_{i,t-1} \Gamma_\beta + \nu_{i,t-1}, \quad (\text{A.2})$$

where z is a vector of observable characteristics. The Γ_β matrix maps from L characteristics to K latent factors and is simultaneously estimated with f_t using alternating least squares.²⁵ Following Kelly, Moskowitz, and Pruitt (2021), the final model obtains out-of-sample conditional expected returns from betas and rolling averages of latent factors as follows

$$\begin{aligned} \mathbb{E}_{t-1} [r_{i,t}] &= \hat{\beta}_{i,t-1} \lambda_{t-1}, \\ \text{where } \lambda_{t-1} &\equiv \mathbb{E}_{t-1} [\hat{f}_t] = \frac{1}{t-1} \sum_{\tau=1}^{t-1} \hat{f}_\tau. \end{aligned} \quad (\text{A.3})$$

I estimate IPCA using a large collection of observable characteristics. The first set of variables is borrowed from the literature on cross-sectional stock return predictability. These include 209 characteristics obtained from Chen and Zimmermann (2022). I filter out stock characteristics for which more than a quarter of observations are missing, leaving me with a total of 130 variables. The second set is a collection of 21 option characteristics commonly used to predict option returns or underlying stock returns. These variables require manual construction, which is detailed in Appendix A.1.

As common in the IPCA literature, I rank-normalize characteristics into the interval $[-0.5, +0.5]$. I proceed as follows. First, I compute cross-sectional ranks for each characteristic. Second, I divide each rank by the number of nonmissing observations at each time and subtract 0.5. Third, I replace missing values with zero, i.e. the cross-sectional median during that month (e.g. as in Gu, Kelly, and Xiu, 2021).

I evaluate model fits using the R^2 statistics proposed in Kelly et al. (2019). The first measure describes how well betas *and* systematic risk factors describe the realized returns of options. It is

²⁵I use the code provided on Seth Pruitt's website: <https://sethpruitt.net/research>.

defined as

$$\text{Total } R^2 = 1 - \frac{\sum_{i,t} (r_{i,t} - \hat{\beta}'_{i,t-1} \hat{f}_t)^2}{\sum_{i,t} r_{i,t}^2}. \quad (\text{A.4})$$

The second measure aims to summarize the model's ability to explain realized returns based on an option's riskiness. It is given as

$$\text{Predictive } R^2 = 1 - \frac{\sum_{i,t} (r_{i,t} - \hat{\beta}'_{i,t-1} \lambda_{t-1})^2}{\sum_{i,t} r_{i,t}^2}, \quad (\text{A.5})$$

where $\hat{\beta}'_{i,t-1} \lambda_{t-1}$ is the conditional expected return.

A.3. TABLES AND FIGURES OMITTED FROM THE MAIN TEXT

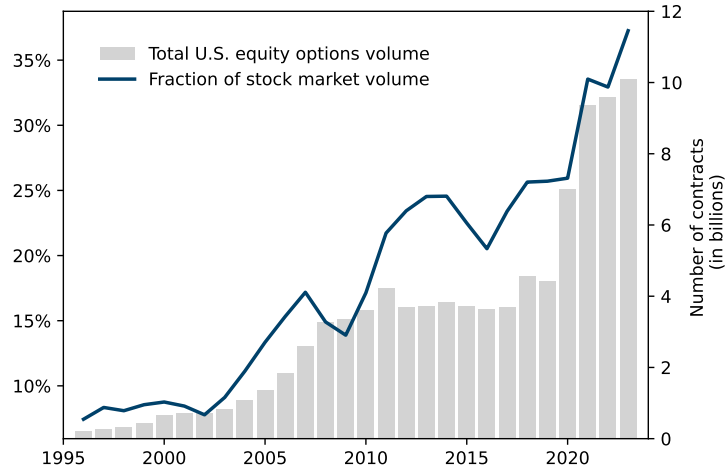


Figure A.1: Trading volume in U.S. single stock options. The figure displays the annual number of single-name equity option contracts traded across all U.S. option exchanges (grey bars). It further shows the percentage share of equity options volume to total U.S. stock market volume. Stock market volume includes optionable as well as non-optionable stocks. Annual data from 1996 to 2023 is from CRSP and the Options Clearing Corporation (OCC).

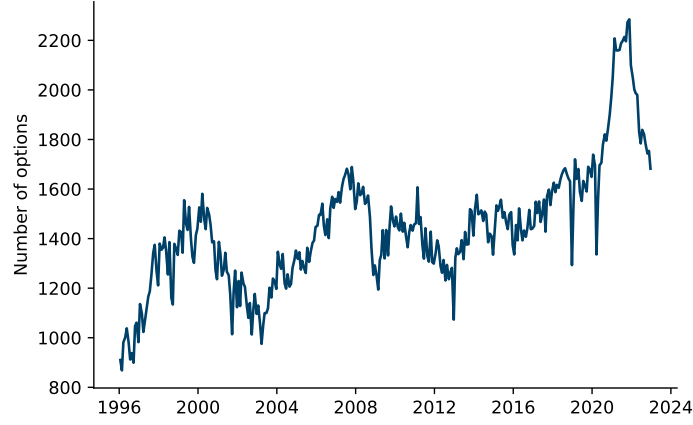
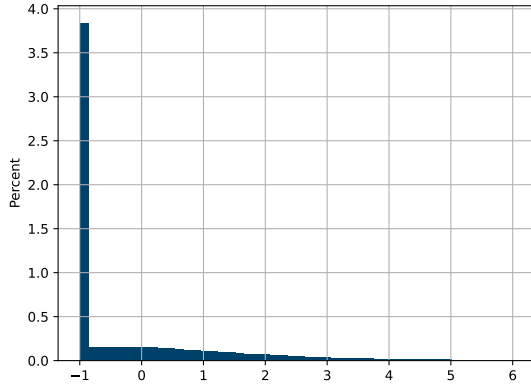
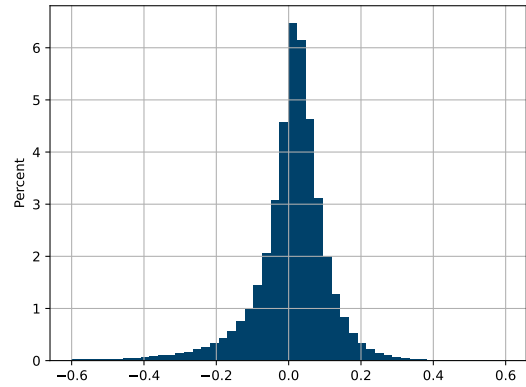


Figure A.2: Evolution of sample size. The figure displays the number of at-the-money call options surviving after applying standard data filters. The monthly sample is from 1996 to 2022.



(a) Raw call return



(b) Delta-hedged call return

Figure A.3: Return distributions. The figure shows pooled distributions of monthly raw (Panel a) and delta-hedged (Panel b) call returns. Each sample covers 436,512 observations from 1996 to 2022.

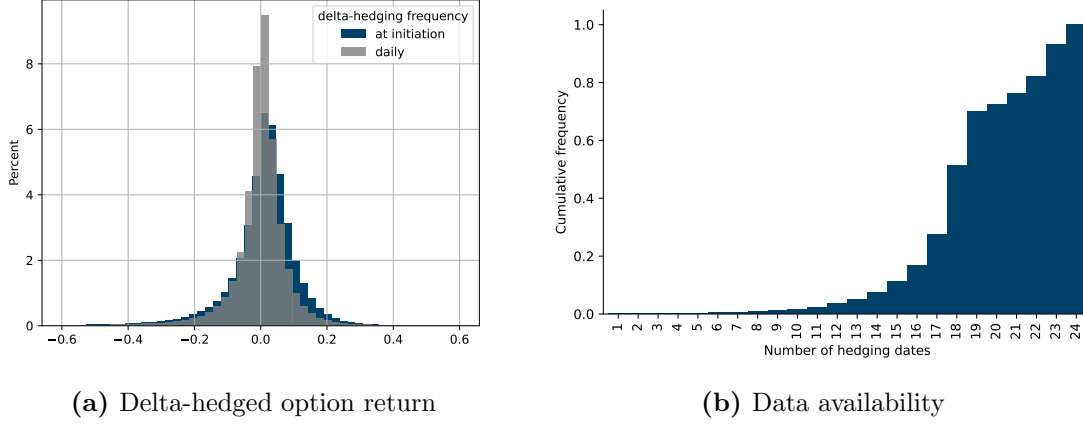


Figure A.4: Daily delta-hedging. The figure shows the cumulative distribution of the number of re-hedging days within a month (Panel a) and the distribution of call returns with delta-hedging at the daily frequency compared to with delta-hedging at initiation only (Panel b). Data is monthly from January 1996 to December 2022.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\hat{r}_{t+1}^{CNN}$		1.93*** (8.54)		1.53*** (7.21)		2.01*** (8.74)		1.53*** (6.86)	1.61*** (7.72)
VRP_{1M}^{MF}	0.57* (1.74)	0.17 (0.51)							-0.4 (-0.62)
VRP_{12M}^{MF}			1.23*** (3.68)	0.76** (2.18)					0.29 (0.56)
VRP_{1M}^{CRR}					0.22 (0.84)	-0.04 (-0.14)			0.06 (0.12)
VRP_{12M}^{CRR}							1.32*** (3.66)	0.94*** (2.52)	0.78 (1.53)
Adj. R^2	1.66	3.19	2.13	3.41	1.17	2.8	1.86	3.21	5.93

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table A.1: Cross-sectional regressions to predict option returns with the VRP. The table shows average slopes ($\times 100$) and t-statistics (in parentheses) from Fama and MacBeth cross-sectional regressions of delta-hedged option returns on CNN predictions and various measures of the VRP. VRP_{1M}^{MF} (VRP_{12M}^{MF}) is the difference between model-free implied variance and historical variance from daily stock returns over the most recent month (12 months). VRP_{1M}^{CRR} (VRP_{12M}^{CRR}) is the average of the at-the-money call and put implied volatility minus the historical volatility from daily stock returns over the most recent month (12 months). The table further reports the average adjusted R^2 in percent. I estimate regressions using weighted-least squares with dollar open interest as the weight. Independent variables are first standardized by their cross-sectional rank and then scaled to the interval $[0, 1]$. t-statistics are adjusted for heteroscedasticity and autocorrelation (Newey and West, 1987). Data are at the monthly frequency from January 2003 to December 2022.

	(1)	(2)	(3)	(4)	(5)	(6)
$\hat{r}_{t+1}^{CNN}$		1.55*** (5.29)		1.44*** (5.09)		1.39*** (5.48)
Beta	-0.46 (-1.59)	-0.53* (-1.83)			-0.4 (-1.32)	-0.49 (-1.57)
Size	-0.14 (-0.36)	-0.14 (-0.36)			-0.36 (-0.95)	-0.38 (-1.0)
BM	-0.32 (-1.47)	-0.34 (-1.58)			-0.19 (-0.91)	-0.2 (-1.0)
OperProf	0.01 (0.04)	-0.03 (-0.14)			-0.02 (-0.08)	-0.06 (-0.27)
AssetGrowth	0.6*** (2.78)	0.51*** (2.41)			0.48** (2.09)	0.42* (1.84)
Mom12m	-0.31 (-1.07)	-0.35 (-1.23)			-0.25 (-0.85)	-0.31 (-1.02)
IdioVolAHT	-0.85** (-2.03)	-0.47 (-1.14)			-0.67 (-1.6)	-0.37 (-0.89)
Cash			0.54* (1.82)	0.6** (2.09)	0.55* (1.85)	0.6** (2.07)
VarCF			-0.1 (-0.33)	0.09 (0.34)	0.16 (0.53)	0.17 (0.58)
ForecastDispersion			-0.15 (-0.57)	0.06 (0.26)	0.03 (0.13)	0.09 (0.4)
ShareIss1Y			0.22 (0.95)	0.22 (0.98)	0.08 (0.37)	0.13 (0.59)
ShareIss5Y			0.02 (0.06)	0.04 (0.15)	-0.12 (-0.53)	-0.13 (-0.6)
Adj. R^2	16.56	17.98	12.23	14.03	22.68	23.8

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table A.2: Cross-sectional regressions to predict option returns from underlying stock fundamentals. The table shows average slopes ($\times 100$) and t-statistics (in parentheses) from [Fama and MacBeth](#) cross-sectional regressions of delta-hedged option returns on CNN predictions and a set of control variables. The table further reports the average adjusted R^2 in percent. I estimate regressions using weighted-least squares with dollar open interest as the weight. Independent variables are first standardized by their cross-sectional rank and then scaled to the interval $[0, 1]$. t-statistics are adjusted for heteroscedasticity and autocorrelation ([Newey and West, 1987](#)). Stock characteristics are sourced from [Chen and Zimmermann \(2022\)](#). Data are at the monthly frequency from January 2003 to December 2022.

	dep = $\hat{r}_{t+1}^{\text{CNN}}$			
	(1)	(2)	(3)	(4)
Var ^Q	12.45*** (10.83)			2.10** (2.21)
Skew ^Q	-0.82*** (-7.34)			-0.53*** (-4.29)
Kurt ^Q	-0.20*** (-3.03)			-0.05 (-1.30)
DownVar ^Q	-19.18*** (-10.76)			-6.22*** (-3.48)
TsVar ^Q	2.84*** (5.93)			0.55 (1.09)
TsSkew ^Q	0.35*** (3.65)			0.28*** (2.76)
TsKurt ^Q	0.09 (1.41)			0.03 (0.91)
IVolATM		0.37* (1.78)		2.04*** (3.22)
C-P Spread		5.89*** (7.89)		5.21*** (12.84)
IVSlope		-2.28*** (-3.18)		-2.86*** (-4.01)
IVSkew		-2.36*** (-5.02)		-0.81* (-1.93)
VoV		0.09 (0.32)		-0.22 (-1.25)
OS			0.19* (1.74)	0.04 (0.70)
ODP			-0.01*** (-3.13)	-0.01*** (-3.71)
Var ^P			0.37** (2.32)	1.08** (2.27)
VRP			0.16 (0.79)	1.02** (2.13)
Adj. R^2	36.66	33.12	24.33	48.46

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table A.3: Contemporary cross-sectional regressions to explain CNN predictions. The table shows average slopes ($\times 100$) and t-statistics (in parentheses) from contemporary cross-sectional regressions of CNN predictions on a set of option characteristics. Variable definitions are provided in Appendix A.1. I estimate regressions using weighted-least squares with dollar open interest as the weight. Independent variables are first standardized by their cross-sectional rank and then scaled to the interval $[0, 1]$. t-statistics are adjusted for heteroscedasticity and autocorrelation (Newey and West, 1987). Data are at the monthly frequency from January 2003 to December 2022.

(a) Stock characteristics						
	K					
	1	2	3	4	5	6
<i>Panel A: Individual options</i>						
Total R^2	9.58	10.63	11.23	11.67	11.91	12.21
Pred. R^2	0.02	0.09	0.20	0.14	0.14	0.15
<i>Panel B: Managed portfolios</i>						
Total R^2	91.01	93.55	95.04	95.87	96.17	96.49
Pred. R^2	0.12	0.43	0.65	-0.15	0.22	0.17
<i>Panel C: Out-of-sample annualized Sharpe ratios</i>						
Sharpe ratio (EW)	0.31	0.53	1.65	1.89	2.21	2.37
Sharpe ratio (OIW)	0.07	0.18	0.43	0.55	0.46	0.55

(b) Option characteristics						
	K					
	1	2	3	4	5	6
<i>Panel A: Individual options</i>						
Total R^2	9.87	10.80	11.40	11.75	11.93	12.05
Pred. R^2	0.04	0.31	0.34	0.37	0.38	0.39
<i>Panel B: Managed portfolios</i>						
Total R^2	92.59	95.68	97.78	98.28	98.52	98.63
Pred. R^2	0.01	0.55	0.96	0.93	0.94	0.96
<i>Panel C: Out-of-sample annualized Sharpe ratios</i>						
Sharpe ratio (EW)	0.99	2.87	3.29	3.55	3.58	3.59
Sharpe ratio (OIW)	0.03	1.18	1.00	0.98	0.93	0.97

(c) Stock and option characteristics						
	K					
	1	2	3	4	5	6
<i>Panel A: Individual options</i>						
Total R^2	9.75	10.58	11.50	11.98	12.24	12.54
Pred. R^2	0.03	0.12	0.35	0.44	0.41	0.39
<i>Panel B: Managed portfolios</i>						
Total R^2	89.20	90.94	93.36	94.90	95.28	95.67
Pred. R^2	0.11	-0.63	0.73	1.64	1.53	1.35
<i>Panel C: Out-of-sample annualized Sharpe ratios</i>						
Sharpe ratio (EW)	0.46	3.22	3.82	3.99	3.91	3.96
Sharpe ratio (OIW)	0.06	1.28	1.22	1.08	1.07	1.12

Table A.4: IPCA fits. This table shows IPCA fits for different information sets. Panels A and B show total R^2 and predictive R^2 of the IPCA model (in percent). Panel A (B) reports results for individual options (managed portfolios). Panel C displays annualized Sharpe ratios of long-short portfolios sorted on conditional expected returns. EW (OIW) indicate equal (dollar open interest) weighted portfolio returns. IPCA is estimated out-of-sample using an expanding window with a minimum of 7 years of data. The monthly sample is from January 2003 to December 2022.

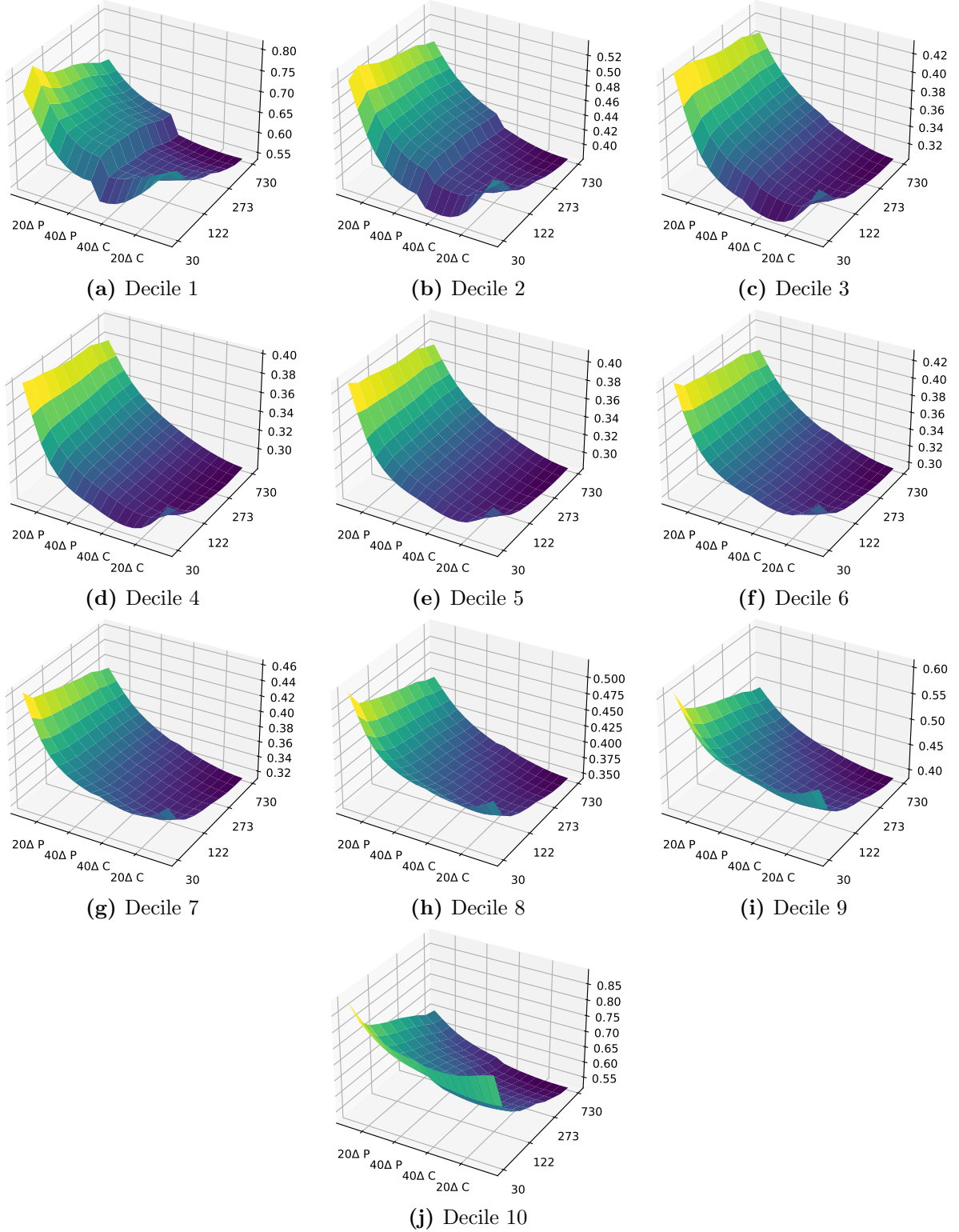


Figure A.5: Average volatility surfaces of CNN decile portfolios. This figure shows the average volatility surface for options within each CNN decile portfolio. First, at each portfolio formation date and separately for each decile portfolio, I calculate the dollar open interest weighted average of each individual value on the volatility surface. I then take the time-series averages of these monthly means for each decile portfolio and implied volatility. The sample period is monthly and spans January 2003 to December 2022.